

Six Functors and Gestalten

Talk 10 — Gestalten Seminar

after [Sch26], Lecture IX (*up to the extension to stacks*)

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Goal of the talk

Starting from a (presentable) six-functor formalism $D: \text{Span}(C) \rightarrow 1\text{Pr}$, we construct *canonically* a functor

$$[-]_D: C \longrightarrow \text{Gest} := \text{StRing}^{\text{op}}, \quad X \longmapsto [X]_D = \text{Gest}(A_{D,X}),$$

the *transmutation* of D , through which D factors as “geometry + quasicoherent sheaves”. The engine is the *category of kernels* $\mathbb{K}_D$; iterating it produces the Stefanich ring $A_{D,X}$. We show $[-]_D$ commutes with finite limits, that every $[f]_D$ is 1-étale and 1-proper, and that $\mathcal{O}(-)_1 \circ [-]_D$ recovers the $(f^*, \otimes)$ -part of D (Theorem 3.4). We stop just before the extension to stacks, stating it without proof in the closing outlook.

Conventions

Categories. Throughout, “category” means ∞ -category, and C is a small one with finite limits.

The formalism. $D: \text{Span}(C) \rightarrow 1\text{Pr}$ is a *presentable six-functor formalism* — a lax symmetric monoidal functor on the span category — where 1Pr denotes presentable categories and $2\text{Pr} = \text{Mod}_{1\text{Pr}}(1\text{Pr})$. Following [Sch26] we fix the cardinal $\kappa = \aleph_1$ throughout, so $1\text{Pr} = \text{Pr}_{\aleph_1}^{\text{L}}$ and $n\text{Pr} = \text{Mod}_{(n-1)\text{Pr}}(1\text{Pr})$; “presentable” means $\aleph_1$ -presentable, and the “countable colimit” and “countably presented” hypotheses invoked below are taken with respect to this κ .

Kernels and Gestalten. $\mathbb{K}_D$ is the *category of kernels* (Section A) and $\text{Gest} = \text{StRing}^{\text{op}}$. We write $\mathbb{1}$ for the monoidal unit, and $\mathcal{O}(-)_1$ (also written D_{qc}) for “global sections / quasicoherent sheaves” — the bottom floor of a Stefanich ring.

D-properties. A morphism is *D-suave*, *D-prim*, *D-smooth*, *D-étale*, or *D-proper* when it has the corresponding property for D (Section A); the unprefix adjectives (smooth, étale, proper, ...) denote the underlying geometric notions.

1 From motives to six-functor formalisms

In algebraic geometry there are many cohomology theories: the ℓ -adic ones, one for each prime ℓ ; crystalline cohomology; de Rham and Betti cohomology. Grothendieck envisioned a *universal* system of coefficients that would dictate all the rest — an initial cohomology theory out of which every other one is read off. In its original, abelian-categorical form, the idea asks for a category Mot_k of *motives*, together with an assignment $X \mapsto \text{Mot}(X)$ (the *motive* of X), such that every reasonable “cohomology functor” $h^\bullet: \text{Sch}_k^{\text{op}} \rightarrow \mathbb{A}$ valued in an abelian category $\mathbb{A}$ factors through it:

$$\begin{array}{ccc} \text{Sch}_k^{\text{op}} & \xrightarrow{\text{Mot}} & \text{Mot}_k \\ & \searrow h^\bullet & \downarrow \\ & & \mathbb{A}. \end{array}$$

The dashed leg is a *realization*: each Weil cohomology — Betti, de Rham, ℓ -adic — is one realization of the universal motive. All the variation *among* cohomology theories is carried by this choice of realization, while the geometric input $X \mapsto \text{Mot}(X)$ stays fixed. This shape of factorization — a fixed universal object, recovered into each concrete theory by a single leg — is the one we will keep meeting; the goal of the talk is its modern, categorified incarnation.

The first move is to categorify the coefficients. Studying a category C of geometric objects — say schemes of finite type over $\mathbb{Z}$ — on its own is hard; one probes it through linear invariants. The bare association $X \mapsto H^\bullet(X)$ is coarse, since many non-isomorphic spaces share the same cohomology. One remembers far more by attaching to each X a whole category of *coefficients*: *local systems* $\mathcal{L}$, which themselves carry cohomology $H^\bullet(X, \mathcal{L})$ and are transported along a map $f: Y \rightarrow X$ by fibrewise cohomology $f_*\mathcal{L}$. To allow arbitrary maps, not just smooth proper ones, one enlarges these into a category of *sheaves*. Cohomology then acquires a dual companion, *cohomology with compact support* $H_c^\bullet$, already visible in Poincaré duality.

The structure that packages all of this is a *six-functor formalism* (defined carefully in Section A). Informally, it assigns to each $X \in C$ a symmetric monoidal *category of coefficients* $D(X)$ — in our setting $D(X) \in \mathbf{1Pr}$ — and to each map $f: Y \rightarrow X$ two adjoint pairs

$$f^* \dashv f_*, \quad f_! \dashv f^!,$$

encoding pullback, pushforward (cohomology), exceptional pushforward (cohomology with compact support), and exceptional pullback, all tied to the tensor product $\otimes$ by base change and a projection formula.

We now look for the categorified analogue of Mot_k : a fixed target through which *every* six-functor formalism factors. A first candidate is the category AnStck of *analytic stacks*, through which the $(f^*, \otimes)$ -part $D^{*, \otimes}$ often factors. There is a suggestive — if imperfect — analogy with Grothendieck’s motives. Both factor a whole family of theories through a single universal object, but of *different kinds* and by holding opposite legs of the triangle fixed: Grothendieck fixes the geometric objects

and varies the coefficients, motives Mot_k being the universal *coefficient*; Scholze fixes the coefficient target and varies the geometric object, AnStck (later Gest) being the universal *geometric object*:

	Grothendieck: motives	Scholze: six functors
factorization	$\begin{array}{ccc} \text{Sch}_k^{\text{op}} & \xrightarrow{\text{Mot}} & \text{Mot}_k \\ & \searrow h^\bullet & \downarrow \text{real.} \\ & & \mathbb{A} \end{array}$	$\begin{array}{ccc} C^{\text{op}} & \xrightarrow{X \mapsto X^?} & \text{AnStck}^{\text{op}} \\ & \searrow D^{*, \otimes} & \downarrow D_{qc} \\ & & \text{CAlg}(\text{1Pr}) \end{array}$
universal object	Mot_k — <i>coefficient</i>	AnStck (later Gest) — <i>geometric object</i>
front leg	Mot — <i>fixed</i>	transmutation $X \mapsto X^?$ — <i>varies with D</i>
recovering leg	realization $\text{Mot}_k \rightarrow \mathbb{A}$ — <i>varies</i>	$D_{qc} = \mathcal{O}(-)_1$ — <i>fixed</i>
variation lives in	the realization	the transmutation

The same triangle, with the fixed and varying legs interchanged. Scholze packages all the realizations into the single theory-independent leg $\mathcal{O}(-)_1 = D_{qc}$ — the same for every D , the leg that reads the $(f^*, \otimes)$ -part of D back off the transmutation (Theorem 3.4) — and studies the front leg instead: the *transmutation* $X \mapsto X^?$, after Bhatt [Bha22, Remark 2.3.8], which realizes a geometric object inside the universal category of geometric objects and commutes with finite limits. So the classical Betti and de Rham realizations are just $\mathcal{O}(-)_1$ applied to the transmutations by the Betti and de Rham ring stacks. The one caveat is the dashed top leg: AnStck achieves this only for *some* D .

Before turning to the obstruction, we record when the dashed leg *does* exist. For schemes (separated of finite type over a field, say) such transmutations are produced from a *ring stack*, by the following criterion of Scholze.

Theorem 1.1 ([Sch25, Theorem 10.6]). *Work in the category of analytic stacks over some base analytic stack S . Let k be a ring and let R be a k -algebra stack satisfying the following conditions.*

- (1) *The map $f: R \rightarrow *$ is $!$ -able and D -smooth, and the map $\mathbb{1} \rightarrow f_! f^! \mathbb{1}$ is an equivalence.*
- (2) *The origin $i: * \xrightarrow{0} R$ is a closed immersion, the units $j: R^\times \hookrightarrow R$ yield an open immersion,¹ and*

$$j_! \mathbb{1} \rightarrow \mathbb{1} \rightarrow i_* \mathbb{1}$$

is a fibre sequence in $D_{qc}(R)$.

This induces a transmutation functor

$$X \mapsto X_R$$

¹Implicit here is that the composite is zero, i.e. $R^\times$ and the origin 0 have empty intersection.

taking any separated scheme X of finite type over k to the 0-truncated analytic stack over S given by the sheafification of the functor

$$\text{AnRing}_S \ni A \mapsto X(R(A)).$$

The functor $X \mapsto X_R$ commutes with finite limits and Zariski gluing. Moreover, the functor $X \mapsto X_R$ sends étale maps to D-étale maps, proper maps to D-proper maps, and smooth maps to D-smooth maps, and thus $X \mapsto D_{qc}(X_R)$ yields a six-functor formalism with these properties.

For instance, the Betti stack and the solid de Rham stack arise from such ring stacks.

The fixed second leg D_{qc} satisfies the *categorical Künneth formula*: by [Kes25], on analytic stacks over an affine base the external product is an equivalence.² Hence, if D factors as above, then D itself satisfies categorical Künneth:

$$D(X) \otimes_{D(*)} D(Y) \xrightarrow{\sim} D(X \times Y) \quad \text{for all } X, Y.$$

For SH, for IndCoh, and for ℓ -adic sheaves, however, the categorical Künneth formula *fails* — so none of these factors through AnStck in this way.

Remark 1.2 (Why the Künneth formula fails for SH, IndCoh, and ℓ -adic sheaves). In each case the external product $D(X) \otimes_{D(*)} D(Y) \xrightarrow{\boxtimes} D(X \times Y)$ fails to be an equivalence, so the kernel category $\mathbb{K}_D$ is strictly larger than $\text{Mod}_{D(*)}(\text{1Pr})$ (Proposition 2.3). (For ℓ -adic sheaves the external product is moreover fully faithful [Gai25], so there the failure is one of essential surjectivity; for IndCoh and SH the cited sources assert only non-equivalence.)

- *ℓ -adic sheaves* [Gai25]. The functor $\text{Shv}(Y) \otimes \text{Shv}(Y') \xrightarrow{\boxtimes} \text{Shv}(Y \times Y')$ is *never* an equivalence once Y, Y' are positive-dimensional of finite type. For a witness take $Y' = Y$ *smooth* (so that $T^*(Y \times Y)$ and singular support are available) and the diagonal $\Delta: Y \hookrightarrow Y \times Y$: the constructible sheaf $\Delta_* \mathbb{1}_Y$ is compact, with singular support the conormal $N_\Delta^* \subset T^*(Y \times Y)$. Every external product satisfies $\text{SS}(F \boxtimes G) \subseteq \text{SS}(F) \times \text{SS}(G)$ [KS90, §5.4]. But N_Δ^* is not contained in any finite union of products of conic Lagrangians: under $N_\Delta^* \cong T^*Y$, a point $(y, \xi, y, -\xi) \in \Lambda \times \Lambda'$ forces $(y, \xi) \in \Lambda \cap a(\Lambda')$ (a the fibrewise antipode), a locus of dimension at most $d = \dim Y$, while N_Δ^* has dimension $2d$. Since $\boxtimes$ is fully faithful and preserves colimits, any preimage of the compact object $\Delta_* \mathbb{1}_Y$ would itself be compact — hence a retract of a finite colimit of external products $F \boxtimes G$ — which the singular-support bound forbids. So $\Delta_* \mathbb{1}_Y$ escapes the image. (We argue with the topological singular support of [KS90, §5.4]; for ℓ -adic coefficients the corresponding theory is Beilinson’s [Bei16].)

²Precisely, [Kes25, Cor. 1.5.1]: over a base with representable diagonal admitting an affine $!$ -cover — in particular over an affine base — every $!$ -able object is Künneth. Over a general stacky base the *relative* formula may fail, but that is not what is at stake here.

- *Ind-coherent sheaves.* The obstruction sits at the bottom (1Pr) level, and is exactly the failure of Künneth: over a base S the formalism $X \mapsto \mathrm{IndCoh}(X)$ takes values in $1\mathrm{Pr}_S = \mathrm{Mod}_{\mathrm{QCoh}(S)}(1\mathrm{Pr})$, the 2-category of $\mathrm{QCoh}(S)$ -linear presentable categories (= 2- $\mathrm{QCoh}(S)$), and its right-lax monoidal structure is the external product

$$\mathrm{IndCoh}(X_1) \otimes_{\mathrm{QCoh}(S)} \mathrm{IndCoh}(X_2) \longrightarrow \mathrm{IndCoh}(X_1 \times_S X_2).$$

As Gaitsgory–Rozenblyum–Varshavsky note [GRV25], even when S, X_1, X_2 are smooth as schemes (not necessarily smooth over S) this is *not* an equivalence as soon as $X_1 \times_S X_2$ fails to be smooth, since IndCoh agrees with QCoh only on smooth targets. Hence Stefanich’s 2- $\mathrm{IndCoh}(S)$ [Ste25] properly enlarges $1\mathrm{Pr}_S$. (Contrast QCoh , which *does* satisfy Künneth on perfect stacks [BFN10, Thm. 4.14].)

- *Motivic spectra* [Aok26]. Aoki shows the kernel 2-category $\mathbb{K}_{\mathrm{SH}}$ of SH (his $2\mathrm{SH}(\mathbb{Z})$) is, as a stable presentably symmetric monoidal 2-category, freely generated by a ring stack that is homologically trivial, smooth, and sutured — and is emphatically not the unit, giving a moduli description of the genuinely 2-categorical gestalt of SH . Were Künneth to hold, $\mathbb{K}_{\mathrm{SH}}$ would collapse to the single floor $\mathrm{Mod}_{\mathrm{SH}(\ast)}(1\mathrm{Pr})$.

These three failures are not defects to be repaired — the three theories are perfectly good six-functor formalisms as they stand, and one does not expect categorical Künneth to hold for D at the symmetric-monoidal level — but they are exactly the obstruction to factoring through AnStck , and the construction below does not remove them so much as *relocate* them, level by level, into honest higher-categorical data. We enlarge the rigid target AnStck to the category of *Gestalten* $\mathrm{Gest} = \mathrm{StRing}^{\mathrm{op}}$, and factor *every* presentable six-functor formalism D through a canonically associated transmutation:

$$\begin{array}{ccc} C^{\mathrm{op}} & \xrightarrow{[-]_D} & \mathrm{Gest}^{\mathrm{op}} \\ & \searrow D^*, \otimes & \downarrow \mathcal{O}(-)_1 \\ & & \mathrm{CAlg}(1\mathrm{Pr}). \end{array}$$

In contrast with the analytic-stack picture, which applies only to *some* D , this construction — following Scholze [Sch26] and Stefanich [Ste25] — works for *every* presentable D . We write its front leg $[-]_D$: this is the generic $X \mapsto X^?$ of the table, now produced canonically for every D rather than from an ad hoc ring stack R . *Gestalten* is in this sense the *universal category of geometric objects carrying a six-functor formalism*, the categorified counterpart of Grothendieck’s Mot_k (read with the grain of salt that only the $(f^*, \otimes)$ -part of D is recovered; see Warning 3.5). Quasicoherent sheaves on Gest — the fixed functor $\mathcal{O}(-)_1$ — is the *packaged realization* through which every 1Pr-valued D factors. The price is that $[X]_D$ is now a genuinely higher-categorical object, obtained by iterating the category of kernels $\mathbb{K}_D$ into a *Stefanich ring* $A_{D,X}$. The very failure of Künneth that obstructed the factorization through AnStck is not repaired here but *relocated* — absorbed, level by

level, into the higher target, so that it is the *transmutation* $[-]_{\mathbf{D}}$, not $\mathbf{D}$ itself, that preserves finite limits (Proposition 2.3; Theorem 3.4).

The whole picture is one *ladder of categorical levels*, the skeleton of the rest of the talk: each invariant of X records data one categorical degree higher than the last — and so supports correspondingly richer algebraic operations — while the transmutation $[-]_{\mathbf{D}}$ records every level at once.

Level	Invariant $X \mapsto$	n -categorical algebra	Available operations
0	$H^\bullet(X)$	cohomology ring (numbers)	$+, \times$
1	$\mathbf{D}(X)$	symmetric monoidal category	limits, colimits, $\otimes$
2	$\mathbb{K}_{\mathbf{D}, X}$	category of kernels	limits, colimits, $\otimes$, adjoints, monads
∞	$[X]_{\mathbf{D}}$	Stefanich ring (a Gestalt)	all of the above, at every level

2 The category of kernels $\mathbb{K}_{\mathbf{D}}$

We now construct the finite-limit-preserving transmutation $[-]_{\mathbf{D}}: C \rightarrow \mathbf{Gest}$ at the heart of the universal-geometric-object picture above. The most naive candidate uses $\mathbf{D}$ directly — send each X to the Gestalt of its coefficient category, i.e. postcompose the $(f^*, \otimes)$ -part $\mathbf{D}^{*, \otimes}$ with the bottom-floor inclusion $\mathbf{CAlg}(\mathbf{1Pr}) \hookrightarrow \mathbf{StRing} = \mathbf{Gest}^{\mathrm{op}}$:

$$C^{\mathrm{op}} \xrightarrow{\mathbf{D}^{*, \otimes}} \mathbf{CAlg}(\mathbf{1Pr}) \longrightarrow \mathbf{StRing} = \mathbf{Gest}^{\mathrm{op}}, \quad X \mapsto \mathbf{Gest}(\mathbf{D}(X)).$$

But this need not preserve pullbacks: the product $X \times Y$ is sent to $\mathbf{Gest}(\mathbf{D}(X \times Y))$, whereas the fibre product of $\mathbf{Gest}(\mathbf{D}(X))$ and $\mathbf{Gest}(\mathbf{D}(Y))$ over $\mathbf{Gest}(\mathbf{D}(*))$ is $\mathbf{Gest}(\mathbf{D}(X) \otimes_{\mathbf{D}(*)} \mathbf{D}(Y))$ — the relative tensor over $\mathbf{D}(*)$ computes the fibre product over $\mathbf{Gest}(\mathbf{D}(*))$ in $\mathbf{Gest} = \mathbf{StRing}^{\mathrm{op}}$ (equivalently the product in the slice $\mathbf{Gest}_{/\mathbf{Gest}(\mathbf{D}(*))}$), not the absolute product. The two coincide exactly when $\mathbf{D}$ satisfies the *categorical Künneth formula* $\mathbf{D}(X) \otimes_{\mathbf{D}(*)} \mathbf{D}(Y) \xrightarrow{\sim} \mathbf{D}(X \times Y)$ — which, as Remark 1.2 showed, generally fails (Proposition 2.3). The *category of kernels* $\mathbb{K}_{\mathbf{D}}$ is the canonical next step, but it acts one categorical degree at a time. Passing from $\mathbf{D}$ to $\mathbb{K}_{\mathbf{D}}$ need not make Künneth hold outright; it *lifts* the obstruction one categorification up, so that the kernel theory $X \mapsto \mathbb{K}_{\mathbf{D}, X}$ may itself fail Künneth at the next level. Iterating the construction into the Stefanich ring (Section 3) then clears the obstruction level by level, yielding a transmutation that genuinely preserves finite limits.

The kernel 2-category is constructed in Section A (Definition A.9); here $\mathbb{K}_{\mathbf{D}}$ is its absolute case, over the terminal object of C , taken in the presentable form of the following definition (the precise comparison with the small enriched 2-category of Definition A.9 is the subject of Aoki’s theorem below).

Definition 2.1. Let C be a category with finite limits. Regarding the six-functor formalism $\mathbf{D}$ as a commutative algebra object in $\mathbf{Fun}(\mathbf{Span}(C), \mathbf{1Pr})$, equipped with the Day convolution symmetric

monoidal structure, the 2-category of kernels of D is the module category

$$\mathbb{K}_D := \text{Mod}_D(\text{Fun}(\text{Span}(C), \text{1Pr})).$$

This comes with a symmetric monoidal functor from $\text{Span}(C)^{\text{op}}$, given as the composite

$$\text{Span}(C)^{\text{op}} \rightarrow \text{Fun}(\text{Span}(C), \text{1Pr}) \rightarrow \text{Mod}_D(\text{Fun}(\text{Span}(C), \text{1Pr})) = \mathbb{K}_D,$$

namely the Yoneda embedding followed by the free D -module functor $D \otimes (-)$. We denote it by $X \mapsto [X]_{\mathbb{K}_D}$. Since $[X]_{\mathbb{K}_D}$ is the free D -module on the representable presheaf at X , it is easy to map out of: by the free-forgetful adjunction composed with the Yoneda lemma, for any $X \in \text{Span}(C)^{\text{op}}$ and $M \in \mathbb{K}_D$ one has

$$\text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, M) = M(X).$$

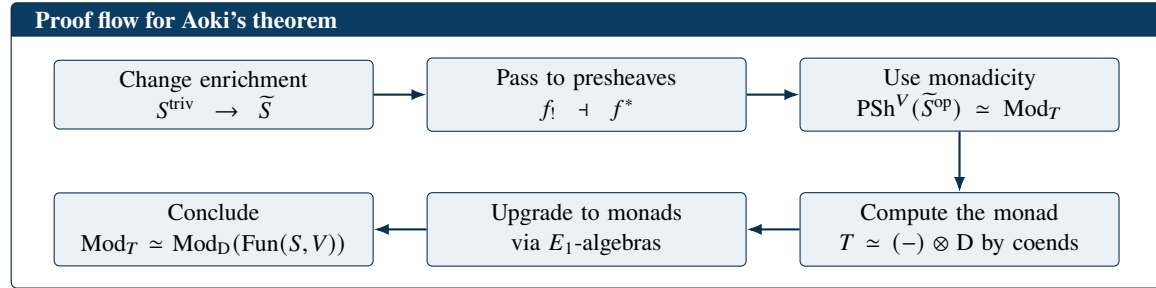
In particular, as the unit object of $\mathbb{K}_D$ is D itself, taking $M = \mathbb{1}$ gives

$$\text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, \mathbb{1}) = D(X).$$

Now every object of $\text{Span}(C)$ is canonically self-dual ([HM24, p. 2.3.9]), and the embedding is symmetric monoidal, so $[Y]_{\mathbb{K}_D} \simeq [Y]_{\mathbb{K}_D}^{\vee}$ and $[X]_{\mathbb{K}_D} \otimes [Y]_{\mathbb{K}_D} \simeq [X \times Y]_{\mathbb{K}_D}$. Hence in general

$$\text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, [Y]_{\mathbb{K}_D}) \simeq \text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D} \otimes [Y]_{\mathbb{K}_D}^{\vee}, \mathbb{1}) \simeq \text{Hom}_{\mathbb{K}_D}([X \times Y]_{\mathbb{K}_D}, \mathbb{1}) = D(X \times Y),$$

which recovers Definition A.9 (there the mapping object is written $\text{Fun}_Y(X_1, X_2) = D(X_1 \times_Y X_2)$; here Y is the terminal object, so $\text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, [Y]_{\mathbb{K}_D})$ is the absolute case $\text{Fun}_*(X, Y)$). The direct computation above matches the two mapping objects on representables; the small enriched kernel 2-category of Definition A.9 (Aoki [Aok26, Def. 2.4]) embeds fully faithfully into $\mathbb{K}_D$ as the representable objects $[X]_{\mathbb{K}_D}$, and the following theorem of Aoki identifies $\mathbb{K}_D$ itself with the 1Pr -enriched presheaves on it.



Theorem 2.2 (Aoki, [Aok26, Theorem 2.6]). *Let S be a small closed symmetric monoidal category and $V \in \text{CAlg}(\text{Pr}^{\text{L}})$. Let $D: S \rightarrow V$ be a lax symmetric monoidal functor. Then there is a canonical equivalence*

$$\text{PSh}^V(\tilde{S}^{\text{op}}) \simeq \text{Mod}_D(\text{Fun}(S, V)),$$

where $\tilde{S}$ is the V -enriched category obtained by base changing the self-enrichment of S along D .

Proof. Aoki credits this argument to H. Heine; we expand it, isolating the monadicity step and computing the resulting monad by transport of structure.

Day convolution and the trivial enrichment. Equip $\text{Fun}(S, V)$ with the Day convolution symmetric monoidal structure, for which a lax symmetric monoidal functor $S \rightarrow V$ is the same datum as a commutative algebra; thus $D \in \text{CAlg}(\text{Fun}(S, V))$. Writing $U: \text{An} \rightarrow V$ for the unit functor — the symmetric monoidal left adjoint with $U(*) = \mathbb{1}_V$ — the pointwise category $\text{Fun}(S, V)$ is the V -presheaf category of S for its *trivial* enrichment S^{triv} ,

$$\text{Fun}(S, V) = \text{PSh}^V((S^{\text{triv}})^{\text{op}}), \quad \underline{\text{Hom}}_{S^{\text{triv}}}(a, b) = U(\text{Hom}_S(a, b)).$$

Change of enrichment. As S is closed it is enriched over itself with mapping objects $\underline{\text{Hom}}_S(a, b)$, and a lax monoidal $G: S \rightarrow V$ base-changes this self-enrichment to the V -enrichment $a, b \mapsto G(\underline{\text{Hom}}_S(a, b))$. Taking $G = U(\text{Hom}_S(\mathbb{1}, -))$ recovers S^{triv} (since $\text{Hom}_S(\mathbb{1}, \underline{\text{Hom}}_S(a, b)) \simeq \text{Hom}_S(a, b)$), while $G = D$ yields $\tilde{S}$, with $\underline{\text{Hom}}_{\tilde{S}}(a, b) = D(\underline{\text{Hom}}_S(a, b))$. Now $U(\text{Hom}_S(\mathbb{1}, -))$ is the unit of $\text{CAlg}(\text{Fun}(S, V))$, so it admits a unique algebra map to D ; base changing along it gives a V -enriched, identity-on-objects functor $S^{\text{triv}} \rightarrow \tilde{S}$, whose opposite we call f . Being a base change of enrichment along a map of commutative algebras, f is symmetric monoidal, and passing to enriched presheaves yields an adjunction

$$f_! \dashv f^* \quad \text{in } \text{Mod}_V(\text{Pr}^{\text{L}}),$$

with $f^*: \text{PSh}^V(\tilde{S}^{\text{op}}) \rightarrow \text{Fun}(S, V)$ the restriction of enrichment (lax monoidal, V -linear) and $f_!$ its V -linear, strong monoidal left adjoint, the enriched left Kan extension. Since $\underline{\text{Hom}}_S(\mathbb{1}, a) \simeq a$, restriction of the monoidal unit recovers D :

$$(f^* \mathbb{1})(a) = \underline{\text{Hom}}_{\tilde{S}}(\mathbb{1}, a) = D(\underline{\text{Hom}}_S(\mathbb{1}, a)) \simeq D(a), \quad \text{i.e. } f^* \mathbb{1} \simeq D \text{ in } \text{CAlg}(\text{Fun}(S, V)).$$

Monadicity. Restriction along an identity-on-objects functor is computed objectwise, so f^* preserves all colimits and is conservative — a map of $\tilde{S}$ -presheaves whose restriction to the (unchanged) collection of objects is an equivalence was already an objectwise equivalence. It is moreover V -linear and admits the left adjoint $f_!$, so the monadicity theorem for presentable module categories of Reutter–Zetto applies: the Eilenberg–Moore comparison is fully faithful [RZ25, Thm. 4.7] and, under these hypotheses, an equivalence [RZ25, Thm. 4.12]. Hence f^* is monadic, and with $T := f^* f_!$ the resulting V -linear monad on $\text{Fun}(S, V)$,

$$\text{PSh}^V(\tilde{S}^{\text{op}}) \simeq \text{Mod}_T(\text{Fun}(S, V)).$$

It remains to identify T — first as an endofunctor, then as a monad.

The underlying endofunctor. Because f is the identity on objects, f^* does not move objects and $T(P)(s') = (f_! P)(s')$; the coend formula for the enriched left Kan extension $f_!$ then gives, for

$P \in \text{Fun}(S, V)$ and $s' \in S$,

$$T(P)(s') \simeq \int^{a \in S} \underline{\text{Hom}}_S(a, s') \otimes P(a) = \int^a D(\underline{\text{Hom}}_S(a, s')) \otimes P(a).$$

On the other hand, Day convolution gives

$$(P \otimes D)(s') = \int^{a, b} \text{Hom}_S(a \otimes b, s') \otimes P(a) \otimes D(b).$$

Integrating over b first (Fubini for coends): for each fixed a , the closed-structure equivalence $\text{Hom}_S(a \otimes b, s') \simeq \text{Hom}_S(b, \underline{\text{Hom}}_S(a, s'))$ followed by the co-Yoneda lemma $\int^b \text{Hom}_S(b, Y) \otimes D(b) \simeq D(Y)$ at $Y = \underline{\text{Hom}}_S(a, s')$ collapses the inner coend to $D(\underline{\text{Hom}}_S(a, s'))$, whence

$$(P \otimes D)(s') \simeq \int^a P(a) \otimes D(\underline{\text{Hom}}_S(a, s')).$$

The two displays agree, naturally in s' , so $T \simeq (-) \otimes D$ as endofunctors of $\text{Fun}(S, V)$ — obtained from the closed structure of S alone, with no rigidity or separate projection formula. (Equivalently, by V -linearity it would suffice to compare the two on representables; the coend does it on all of $\text{Fun}(S, V)$ at once.)

The monad structure, and conclusion. We upgrade $T \simeq (-) \otimes D$ to an equivalence of monads by transport of structure, not by a hand-built comparison (a map of E_1 -algebras is not determined by its underlying map and unit/multiplication compatibilities alone). The computation above, run for a general Y , shows the projection map $A \otimes f^*Y \rightarrow f^*(f_!A \otimes Y)$ is an equivalence — both sides preserve colimits in A and Y , so it suffices to check on representables, where it is the coend identity established above — so f^* is *strong* $\text{Fun}(S, V)$ -linear. With $f_!$ strong monoidal, the adjunction $f_! \dashv f^*$ — hence the monad $T = f^*f_!$ — lives in $\text{Fun}(S, V)$ -modules, so T is an associative algebra in the monoidal category $\text{Fun}_{\text{Fun}(S, V)}^L(\text{Fun}(S, V))$ of colimit-preserving $\text{Fun}(S, V)$ -linear endofunctors. Evaluation at the unit, $G \mapsto G(\mathbb{1})$, is a *monoidal* equivalence

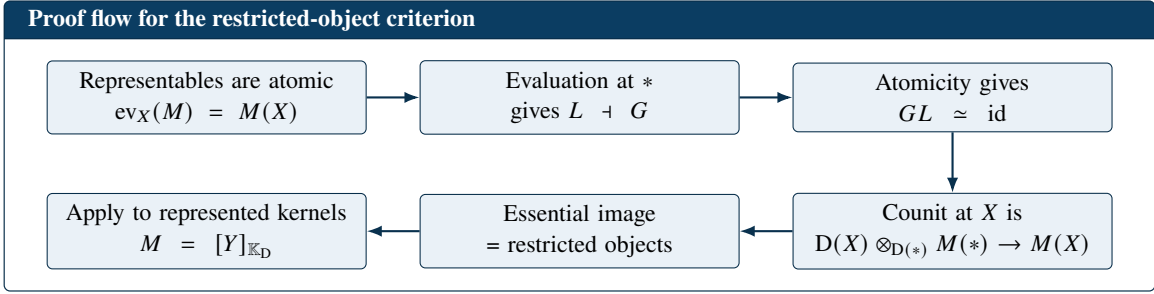
$$\text{Fun}_{\text{Fun}(S, V)}^L(\text{Fun}(S, V)) \xrightarrow{\sim} \text{Fun}(S, V)$$

with inverse $A \mapsto A \otimes (-)$ (the free rank-one module computes its own linear endomorphisms). Applying $\text{Alg}_{E_1}(-)$ — which a monoidal equivalence preserves together with all coherence data — and using $T(\mathbb{1}) \simeq f^*\mathbb{1} \simeq D$, this carries the algebra T to the algebra D . Thus $T \simeq (-) \otimes D$ as monads, the entire coherent identification produced at once. Therefore

$$\text{PSh}^V(\tilde{S}^{\text{op}}) \simeq \text{Mod}_T(\text{Fun}(S, V)) \simeq \text{Mod}_{(-) \otimes D}(\text{Fun}(S, V)) \simeq \text{Mod}_D(\text{Fun}(S, V)),$$

the last step being $\text{Mod}_{(-) \otimes A} \simeq \text{Mod}_A$. Every step is canonical. $\square$

Now, taking $V = \mathbf{1Pr}$ and $S = \text{Span}(C)$, we obtain the desired result.



Proposition 2.3 ([Sch26, Prop. 9.3]). *The natural functor*

$$\mathrm{Mod}_{D(*)}(1\mathrm{Pr}) \longrightarrow \mathbb{K}_D$$

is fully faithful, with essential image the “restricted³” objects: $M \in \mathbb{K}_D$ lies in it iff $D(X) \otimes_{D()} M(*) \rightarrow M(X)$ is an equivalence for all X . In particular,*

$$\mathbb{K}_D = \mathrm{Mod}_{D(*)}(1\mathrm{Pr}) \iff D(X) \otimes_{D(*)} D(Y) \xrightarrow{\sim} D(X \times Y) \text{ for all } X, Y.$$

Proof. We work with the description $\mathbb{K}_D = \mathrm{Mod}_D(\mathrm{Fun}(\mathrm{Span}(C), 1\mathrm{Pr}))$ of Definition 2.1 and the formula $\mathrm{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, M) = M(X)$ established there.

Evaluation is atomic. For each X , that formula says the corepresentable of $[X]_{\mathbb{K}_D}$ is the evaluation functor

$$\mathrm{ev}_X = \mathrm{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, -) : \mathbb{K}_D \longrightarrow 1\mathrm{Pr}, \quad M \longmapsto M(X).$$

Colimits in $\mathbb{K}_D$ are created by the forgetful functor to $\mathrm{Fun}(\mathrm{Span}(C), 1\mathrm{Pr})$, where they are computed pointwise; hence ev_X preserves all colimits, and it is $1\mathrm{Pr}$ -linear. Thus every $[X]_{\mathbb{K}_D}$ is $1\mathrm{Pr}$ -atomic. In particular the unit $\mathbb{1} = [*]_{\mathbb{K}_D}$ is $1\mathrm{Pr}$ -atomic, with $\mathrm{End}_{\mathbb{K}_D}(\mathbb{1}) = \mathrm{ev}_*(\mathbb{1}) = D(*)$.

The natural functor and its right adjoint. Since $\mathrm{ev}_*(M) = M(*)$ is naturally a module over $\mathrm{End}_{\mathbb{K}_D}(\mathbb{1}) = D(*)$, the functor ev_* refines to a colimit-preserving functor

$$G : \mathbb{K}_D \longrightarrow \mathrm{Mod}_{D(*)}(1\mathrm{Pr}), \quad M \longmapsto M(*).$$

The functor of the statement is its left adjoint — induction along $D(*) = \mathrm{End}_{\mathbb{K}_D}(\mathbb{1}) \rightarrow \mathbb{K}_D$,

$$L : \mathrm{Mod}_{D(*)}(1\mathrm{Pr}) \longrightarrow \mathbb{K}_D, \quad N \longmapsto N \otimes_{D(*)} \mathbb{1},$$

the unique $1\mathrm{Pr}$ -linear colimit-preserving functor with $L(D(*)) = \mathbb{1}$; one has $L \dashv G$.

Full faithfulness. L is fully faithful iff the unit $N \rightarrow GL(N)$ of the adjunction is an equivalence for every N . Here

$$GL(N) = (N \otimes_{D(*)} \mathbb{1})(*) \simeq N \otimes_{D(*)} \mathbb{1}(*) = N \otimes_{D(*)} D(*) = N.$$

³in the sense of [Gai25]

The middle equivalence is where atomicity enters: ev_* is colimit-preserving and 1Pr-linear, so it commutes with the relative tensor product $N \otimes_{D(*)} (-)$ (a geometric realization of the bar construction); and $\mathbb{1}(*) = D(*)$. Hence $GL \simeq \text{id}$ and L is fully faithful.

The essential image. As L is fully faithful with right adjoint G , an object $M \in \mathbb{K}_D$ lies in the essential image of L iff the counit $\varepsilon_M: LG(M) \rightarrow M$ is an equivalence. Evaluating $LG(M) = M(*) \otimes_{D(*)} \mathbb{1}$ at X — again using that ev_X commutes with $M(*) \otimes_{D(*)} (-)$, together with $\mathbb{1}(X) = D(X)$ —

$$(LG(M))(X) \simeq M(*) \otimes_{D(*)} \mathbb{1}(X) = D(X) \otimes_{D(*)} M(*),$$

and ε_M at X is precisely the canonical map $D(X) \otimes_{D(*)} M(*) \rightarrow M(X)$. So M is in the image — i.e. *restricted* — iff this map is an equivalence for all X , as claimed.

The Künneth criterion. The restricted objects are closed under colimits and under 1Pr-tensoring: LG and id are both colimit-preserving and 1Pr-linear, so the locus where the natural transformation ε is invertible is closed under colimits and 1Pr-tensoring. Since $\mathbb{K}_D$ is 1Pr-atomically generated by the representables $[Y]_{\mathbb{K}_D}$ — i.e. generated under colimits and 1Pr-tensoring — the functor L is essentially surjective — equivalently an equivalence, i.e. $\mathbb{K}_D = \text{Mod}_{D(*)}(1\text{Pr})$ — iff every $[Y]_{\mathbb{K}_D}$ is restricted. Finally, by the $\text{Hom}_{\mathbb{K}_D}$ -formula,

$$[Y]_{\mathbb{K}_D}(*) = \text{Hom}_{\mathbb{K}_D}(\mathbb{1}, [Y]_{\mathbb{K}_D}) = D(Y), \quad [Y]_{\mathbb{K}_D}(X) = \text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, [Y]_{\mathbb{K}_D}) = D(X \times Y),$$

so $[Y]_{\mathbb{K}_D}$ is restricted iff $D(X) \otimes_{D(*)} D(Y) \rightarrow D(X \times Y)$ is an equivalence for all X . Ranging over Y as well, this is exactly the categorical Künneth formula. $\square$

3 Transmutation

We now construct the transmutation functor $C^{\text{op}} \rightarrow \text{Gest}^{\text{op}} = \text{StRing}$. Note that for any $X \in C$, we can repeat Definition 2.1 with the slice $C_{/X}$, defining

$$\mathbb{K}_{D,X} := \text{Mod}_{D_X}(\text{Fun}(\text{Span}(C_{/X}), 1\text{Pr})) \in \text{CAlg}(2\text{Pr}), \quad D_X := D|_{\text{Span}(C_{/X})}.$$

The pullback functor $\text{Span}(C) \rightarrow \text{Span}(C_{/X})$ induces, by passing to free presentable 2-categories, a symmetric monoidal functor

$$\text{Fun}(\text{Span}(C), 1\text{Pr}) \rightarrow \text{Fun}(\text{Span}(C_{/X}), 1\text{Pr}), \quad F \mapsto F|_{C_{/X}}.$$

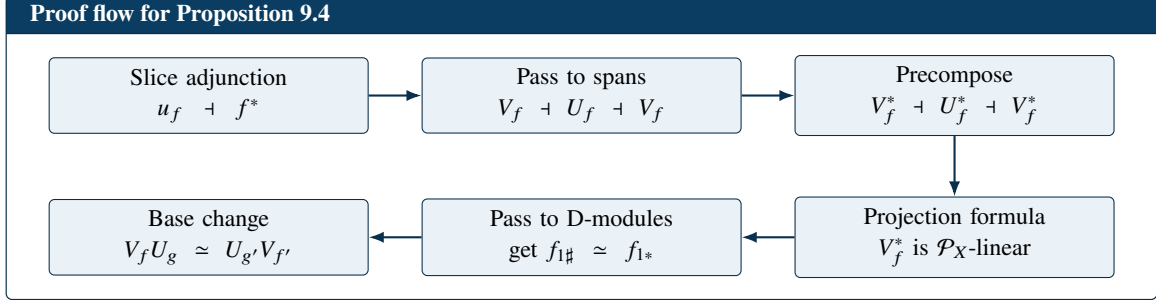
Because D is compatible with this restriction, the two routes to D -modules — restrict then take modules, or take modules then restrict — agree; there is only one sense of “passing to modules over D ”.

For any map $f: Y \rightarrow X$ in C , the pullback functor $f^*: C_{/X} \rightarrow C_{/Y}$ induces a symmetric monoidal functor

$$f_1^*: \mathbb{K}_{D,X} \rightarrow \mathbb{K}_{D,Y},$$

so this assembles into a functor

$$C^{\text{op}} \rightarrow \text{CAlg}(2\text{Pr}).$$



Proposition 3.1 ([Sch26, Prop. 9.4]). *For every morphism $f : Y \rightarrow X$, the functor*

$$f_1^* : \mathbb{K}_{D,X} \longrightarrow \mathbb{K}_{D,Y}$$

admits canonically equivalent left and right adjoints

$$f_{1\sharp} \simeq f_{1*},$$

and their formation is compatible with base change.

Proof. Put

$$S_X := \text{Span}(C_{/X}), \quad \mathcal{P}_X := \text{Fun}(S_X, 1\text{Pr}), \quad \mathbb{K}_{D,X} = \text{Mod}_{D_X}(\mathcal{P}_X),$$

where $\mathcal{P}_X$ is equipped with the Day convolution symmetric monoidal structure and $D_X \in \text{CAlg}(\mathcal{P}_X)$ denotes the restriction of D to the slice over X .

Let

$$u_f : C_{/Y} \longrightarrow C_{/X}, \quad (Z \rightarrow Y) \mapsto (Z \rightarrow Y \xrightarrow{f} X),$$

be postcomposition with f , and let

$$f^* : C_{/X} \longrightarrow C_{/Y}, \quad (W \rightarrow X) \mapsto (W \times_X Y \rightarrow Y),$$

be base change. There is the standard slice adjunction

$$u_f \dashv f^*.$$

Write

$$U_f := \text{Span}(u_f) : S_Y \longrightarrow S_X, \quad V_f := \text{Span}(f^*) : S_X \longrightarrow S_Y.$$

1. *The ambijunction on the span indexing categories.* Equip $C_{/X}$ and $C_{/Y}$ with the unrestricted adequate triples $(-, \text{all}, \text{all})$. The unit and counit of $u_f \dashv f^*$ are

$$\eta_Z^0: Z \longrightarrow Z \times_X Y, \quad \varepsilon_W^0: W \times_X Y \longrightarrow W.$$

Their naturality squares are cartesian. Indeed, for a morphism $W' \rightarrow W$ over X ,

$$W' \times_W (W \times_X Y) \simeq W' \times_X Y,$$

and for a morphism $Z' \rightarrow Z$ over Y , the naturality square of η^0 is the pullback of the graph map $Z \rightarrow Z \times_X Y$.

Consequently, both parts of [Cno26, Cor. 12.2.4] apply. They give

$$U_f \dashv V_f \quad \text{and} \quad V_f \dashv U_f.$$

Thus

$$V_f \dashv U_f \dashv V_f.$$

2. *Passage to the free-presentable functor categories.* Precomposition gives functors

$$U_f^*: \mathcal{P}_X \longrightarrow \mathcal{P}_Y, \quad V_f^*: \mathcal{P}_Y \longrightarrow \mathcal{P}_X.$$

Since precomposition reverses adjunctions [Cno26, Lem. 20.1.6], the two adjunctions above give

$$V_f^* \dashv U_f^* \quad \text{and} \quad U_f^* \dashv V_f^*.$$

Hence already on the free-presentable level one has

$$V_f^* \dashv U_f^* \dashv V_f^*. \tag{1}$$

The functor

$$V_f: S_X \longrightarrow S_Y$$

is symmetric monoidal: it is induced by base change, which preserves finite products in the slice categories. Since $V_f \dashv U_f$, the restriction functor U_f^* may equivalently be identified with left Kan extension along V_f :

$$U_f^* \simeq \text{Lan}_{V_f}.$$

By the universal property of Day convolution [Cno26, Prop. 17.2.13], Lan_{V_f} is symmetric monoidal. Thus U_f^* is a symmetric monoidal functor.

3. *The projection formula.* We claim that for every

$$M \in \mathcal{P}_X, \quad N \in \mathcal{P}_Y,$$

the canonical projection-formula morphism is an equivalence:

$$V_f^* N \otimes M \xrightarrow{\sim} V_f^*(N \otimes U_f^* M). \quad (2)$$

The geometric identity underlying this formula is the following. For $A \in S_X$ and $B \in S_Y$, there is a natural equivalence

$$U_f(B \otimes_Y V_f(A)) \simeq U_f(B) \otimes_X A. \quad (3)$$

Indeed,

$$\begin{aligned} U_f(B \otimes_Y V_f(A)) &= U_f(B \times_Y (A \times_X Y)) \\ &\simeq B \times_X A \\ &= U_f(B) \otimes_X A. \end{aligned}$$

The equivalence is natural in A and B and is compatible with composition of spans, since all the relevant compositions are computed by pullbacks.

Let

$$j_X(A) := \text{Hom}_{S_X}(A, -), \quad j_Y(B) := \text{Hom}_{S_Y}(B, -)$$

denote the covariant representables. Day convolution satisfies

$$j_X(A) \otimes j_X(A') \simeq j_X(A \otimes_X A'),$$

and similarly over Y .

The two indexing adjunctions imply

$$U_f^* j_X(A) \simeq j_Y(V_f A), \quad V_f^* j_Y(B) \simeq j_X(U_f B).$$

Therefore

$$\begin{aligned} V_f^*(j_Y(B) \otimes U_f^* j_X(A)) &\simeq V_f^*(j_Y(B) \otimes j_Y(V_f A)) \\ &\simeq V_f^* j_Y(B \otimes_Y V_f A) \\ &\simeq j_X U_f(B \otimes_Y V_f A) \\ &\simeq j_X(U_f B \otimes_X A) \\ &\simeq j_X(U_f B) \otimes j_X(A) \\ &\simeq V_f^* j_Y(B) \otimes j_X(A). \end{aligned}$$

Thus the projection-formula morphism is an equivalence on representables. Both sides of (2) preserve colimits separately in M and N : Day convolution preserves colimits in each variable, and precomposition is computed pointwise. Since $\mathcal{P}_X$ and $\mathcal{P}_Y$ are the free 1Pr-cocompletions generated by their representables, (2) follows for all M and N .

4. Passage to D-modules. Compatibility of D with restriction to the slices gives an equivalence of commutative algebra objects

$$U_f^* D_X \simeq D_Y. \quad (4)$$

Since U_f^* is symmetric monoidal, it therefore induces

$$f_1^*: \text{Mod}_{D_X}(\mathcal{P}_X) \longrightarrow \text{Mod}_{D_Y}(\mathcal{P}_Y).$$

We now lift V_f^* . Since

$$U_f^* \dashv V_f^*$$

and U_f^* is symmetric monoidal, V_f^* has its canonical lax symmetric monoidal structure. If

$$N \in \text{Mod}_{D_Y}(\mathcal{P}_Y),$$

define a D_X -action on V_f^*N by

$$\begin{aligned} D_X \otimes V_f^*N &\longrightarrow V_f^*(U_f^* D_X \otimes N) \\ &\simeq V_f^*(D_Y \otimes N) \\ &\longrightarrow V_f^*N. \end{aligned} \tag{5}$$

The first arrow is the canonical projection-formula morphism, and the last arrow is induced by the D_Y -action on N . This defines a functor

$$\widetilde{V}_f^*: \mathbb{K}_{D,Y} \longrightarrow \mathbb{K}_{D,X}$$

whose underlying functor is V_f^* .

The standard lifting of a monoidal adjunction to module categories gives

$$f_1^* \dashv \widetilde{V}_f^*. \tag{6}$$

Thus $\widetilde{V}_f^*$ is the right adjoint of f_1^* . Write $f_{1*} := \widetilde{V}_f^*$ for this right adjoint.

5. Ambidexterity: $f_{1\#} \simeq f_{1*}$. Everything up to this point is common to the two routes below; they diverge only in how the *left* adjoint of f_1^* is produced and identified with f_{1*} .

First argument: both underlying adjunctions lift. The projection formula (2) says that V_f^* is strongly $\mathcal{P}_X$ -linear, where $\mathcal{P}_Y$ is regarded as a $\mathcal{P}_X$ -module through U_f^* . Consequently, the other underlying adjunction

$$V_f^* \dashv U_f^*$$

also lifts to the module categories:

$$\widetilde{V}_f^* \dashv f_1^*. \tag{7}$$

The D_X -module structure used in this left-adjoint lift is the same one as in (5), since the projection-formula morphism in (5) is an equivalence.

Hence the single functor $\widetilde{V}_f^*$ is simultaneously a left and a right adjoint of f_1^* :

$$\widetilde{V}_f^* \dashv f_1^* \dashv \widetilde{V}_f^*.$$

Defining

$$f_{1\sharp} := \widetilde{V}_f^* \quad \text{and} \quad f_{1*} := \widetilde{V}_f^*$$

in its two adjoint roles gives the canonical equivalence

$$f_{1\sharp} \simeq f_{1*}.$$

Second argument: the ambidexterity criterion. Name the units and counits of the ambijunction of Step 1: let

$$a: \text{id}_{S_X} \Longrightarrow U_f V_f, \quad b: V_f U_f \Longrightarrow \text{id}_{S_Y}$$

be the unit and counit of $V_f \dashv U_f$, and let

$$\bar{a}: \text{id}_{S_Y} \Longrightarrow V_f U_f, \quad \bar{b}: U_f V_f \Longrightarrow \text{id}_{S_X}$$

be those of $U_f \dashv V_f$. For an arrow $h: A \rightarrow B$ the two spans $A \xleftarrow{\text{id}_A} A \xrightarrow{h} B$ and $B \xleftarrow{h} A \xrightarrow{\text{id}_A} A$ are transposes, hence an adjoint pair in the span 2-category; as the two applications of [Cno26, Cor. 12.2.4] build a, b and $\bar{a}, \bar{b}$ from the same η^0, ε^0 with opposite span orientations, one obtains, componentwise and coherently,

$$a \dashv \bar{b}, \quad b \dashv \bar{a}.$$

Precomposition turns a, b into the unit $\eta_M = M(a)$ and counit $\varepsilon_N = N(b)$ of $U_f^* \dashv V_f^*$; since every $M: S_X \rightarrow 1\text{Pr}$ and $N: S_Y \rightarrow 1\text{Pr}$ carries $a \dashv \bar{b}$ and $b \dashv \bar{a}$ to adjunctions in 1Pr , this unit and counit admit right adjoints $M(\bar{b}), N(\bar{a})$, assembling coherently. Passing through $\text{Mod}_{\mathbb{D}}(-)$ — using, as in (5), that the projection-formula morphism is an equivalence — the unit and counit of $f_1^* \dashv f_{1*}$ likewise admit right adjoints as 2-morphisms in 2Pr .

We may therefore invoke the ambidexterity criterion in 2Pr : if a 1-morphism F has a right adjoint F^R whose unit and counit admit right adjoints, then F^R is also a *left* adjoint of F . Applied to $F = f_1^*$, $F^R = f_{1*}$, it gives $f_{1*} \dashv f_1^*$. On the other hand, left Kan extension along U_f , with the same projection formula, supplies a left adjoint $f_{1\sharp} \dashv f_1^*$; by uniqueness of left adjoints, $f_{1\sharp} \simeq f_{1*}$.

6. *Base change.* Consider a cartesian square

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ f' \downarrow & & \downarrow f \\ X' & \xrightarrow{g} & X. \end{array}$$

There is a canonical equivalence of span functors

$$V_f U_g \simeq U_{g'} V_{f'}. \quad (8)$$

Indeed, for every $W \rightarrow X'$,

$$W \times_X Y \simeq W \times_{X'} Y',$$

naturally in W , and this equivalence is compatible with spans and their composition.

After precomposition, (8) becomes

$$U_g^* V_f^* \simeq V_{f'}^* U_{g'}^*. \quad (9)$$

This equivalence is compatible with the coefficient algebras D and with the projection-formula module structures (5). It therefore lifts to

$$g_1^* \tilde{V}_f^* \simeq \tilde{V}_{f'}^* g_1'^*.$$

Since $\tilde{V}_f^*$ realizes both $f_{1\#}$ and f_{1*} (under the equivalence $f_{1\#} \simeq f_{1*}$ of either argument), the same equivalence gives simultaneously

$$g_1^* f_{1\#} \simeq f_{1\#}' g_1'^*$$

and

$$g_1^* f_{1*} \simeq f_{1*}' g_1'^*.$$

Because both are induced from the single equivalence (9), they agree under the identification $f_{1\#} \simeq f_{1*}$. This proves compatibility with base change. $\square$

This already yields a functor $C \rightarrow \text{Gest}$, $X \mapsto \text{Gest}(\mathbb{K}_{D,X})$, valued in 2-affine Gestalten. Scholze notes here that, being 2-affine, the maps at this stage are 2-proper; in the general formalism one should remember the usual countability caveat for the abstract affine-implies-proper criterion. What we use concretely is the consequence that the higher pieces of primness are already in the affine range, while Proposition 3.1 supplies the remaining kernel-level right adjoint f_{1*} , so the maps are 1-prim. They are not, however, evidently 1-suave: that would need left adjoints for the pullback $\text{Mod}_{\mathbb{K}_{D,X}}(2\text{Pr}) \rightarrow \text{Mod}_{\mathbb{K}_{D,Y}}(2\text{Pr})$ *one level up*, whose existence is not clear from the two-level construction alone. Promoting this stage to 1-proper/1-étale would also require control of diagonals in the transmuted geometry, i.e. finite-limit preservation, which has not yet been proved. These obstructions are what force us to iterate the construction.

Remark 3.2 (Why $\text{Gest}(\mathbb{K}_{D,X})$ is 2-affine). Recall ([Sch26, Def. 1.18]) that a Stefanich ring is a tower $A = (A_1, A_2, \dots)$ with $A_n \in \text{CAlg}(n\text{Pr})$ and $A_n \simeq \text{End}_{A_{n+1}}(\mathbb{1})$, and ([Sch26, Def. 6.5]) that B is *n-affine* when generated by its single component $B_n \in \text{CAlg}(A_{n+1})$ through $B_{m+1} \simeq \text{Mod}_{B_m}(A_{m+1})$ — i.e. it carries no new categorical data above level n . Over the unit base $A_3 = 2\text{Pr}$, $\mathbb{K}_{D,X} = \text{Mod}_{D_X}(\text{Fun}(\text{Span}(C/X), 1\text{Pr})) \in \text{CAlg}(2\text{Pr})$ is exactly such a generator, at level 2; so $\text{Gest}(\mathbb{K}_{D,X})$ is Gest applied to the tower

$$(D(X), \mathbb{K}_{D,X}, \text{Mod}_{\mathbb{K}_{D,X}}(2\text{Pr}), \dots), \quad D(X) = \text{End}_{\mathbb{K}_{D,X}}(\mathbb{1}),$$

whose only genuine datum is $\mathbb{K}_{D,X}$ — so it is 2-affine, the higher analogue of an affine scheme $\mathrm{Spec} R$ with tower $(D(R), \mathrm{Mod}_{D(R)}(1\mathrm{Pr}), \dots)$.

The level is sharp, and reconnects to Proposition 2.3. $\mathrm{Gest}(\mathbb{K}_{D,X})$ would already be 1-affine — that is, $\mathbb{K}_{D,X} \simeq \mathrm{Mod}_{D(X)}(1\mathrm{Pr})$ — precisely when the categorical Künneth formula holds over C/X — explicitly, by Proposition 2.3 applied to the slice C/X (whose terminal object is X , so its $D(*)$ is $D(X)$), when $D(X') \otimes_{D(X)} D(Y') \xrightarrow{\sim} D(X' \times_X Y')$ for all $X', Y' \in C/X$. When it fails, $\mathbb{K}_{D,X} \supsetneq \mathrm{Mod}_{D(X)}(1\mathrm{Pr})$ and $\mathrm{Gest}(\mathbb{K}_{D,X})$ is genuinely 2-affine, a 2-stack morally at the level of $B^2\mathbb{G}_m$. Whether the *full* transmutation $[X]_D = \mathrm{Gest}(A_{D,X})$ is n -affine for some finite n depends on D : this happens exactly when the kernel iteration stabilizes, each higher level contributing genuine data only where Künneth keeps failing. For SH it *does* stabilize, already at level 2: by [Sch26, Prop. 9.9] the gestalt $[\mathrm{Spec} \mathbb{Z}]_{\mathrm{SH}}$ is 2-affine and every projection $[X]_{\mathrm{SH}} \rightarrow [\mathrm{Spec} \mathbb{Z}]_{\mathrm{SH}}$ is 1-affine, so $[X]_{\mathrm{SH}}$ is 2-affine for all X and the tower terminates. By contrast there do exist Gestalten that are n -affine for no finite n — e.g. $B^\infty\mathbb{G}_m$ or the infinite-dimensional affine space [Sch26, Rem. 5.6] — showing the tower need not terminate in general; they are not among the finite-level transmutations we focus on here.

It is worth pausing on *why* iterating helps, since climbing a level might seem only to defer the problem. The kernel construction does not *repair* Künneth — it *relocates* it. By design $\mathrm{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, [Y]_{\mathbb{K}_D}) = D(X \times Y)$: the level-1 discrepancy $D(X \times Y) \neq D(X) \otimes_{D(*)} D(Y)$ is no longer an obstruction but is *absorbed* into the level-2 object $\mathbb{K}_D$, which now knows $D(X \times Y)$ tautologically. What we have bought is a genuinely new six-functor formalism $X \mapsto \mathbb{K}_{D,X}$, valued one categorical degree higher, and it carries its *own* Künneth question $\mathbb{K}_{D,X \times Y} \stackrel{?}{\simeq} \mathbb{K}_{D,X} \otimes_{\mathbb{K}_D} \mathbb{K}_{D,Y}$, which may fail afresh. Each pass thus converts the *failure* of Künneth at one level into honest *data* at the next, and asks the question again one rung up; iterating is relocating the obstruction again and again, and the Stefanich ring $A_{D,X}$ is the fixed point that records the answer at every level at once. For some D the relocation halts — for SH already at level 2 (Remark 3.2) — but the construction is uniform whether or not it does.

Geometrically this is the passage from a ring to a space. The bottom datum $D(X)$ — the level-1 sheaf theory, with functions $\mathcal{O}(X) = \mathrm{End}_{D(X)}(\mathbb{1})$ — does not determine the gestalt $[X]_D$, precisely because Künneth fails; it would, were $[X]_D$ already 1-affine (Proposition 2.3). $\mathrm{Gest}(\mathbb{K}_{D,X})$ is the 2-affine approximation, one layer deeper but still blind above level 2, and the full transmutation $[X]_D = \mathrm{Gest}(A_{D,X})$ is the honest geometric object, assembled from these approximations the way an ∞ -stack is glued from affines: the genuine *space* in Gest attached to X , not the bare algebra $D(X)$ — the level- ∞ row of the ladder, with $[-]_D$ recording every rung.

The failure diagnosed before the remark — that $\mathrm{Gest}(\mathbb{K}_{D,X})$ is only 1-prim, not yet 1-suave or 1-proper — is repaired by climbing one categorical level and repeating the construction. The assignment

$$C^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(2\mathrm{Pr}), \quad X \mapsto \mathbb{K}_{D,X},$$

itself admits, by Proposition 3.1, base-change-compatible left adjoints $f_{1\#} (\simeq f_{1*})$ for every f — and

that base-change compatibility is precisely the base-change axiom one level up — so, assembling these adjoints into a lax symmetric monoidal functor on spans (following [Sch26; Ste25]), it upgrades to a once-categorified six-functor formalism

$$D^{(2)} : \text{Span}(C) \longrightarrow 2\text{Pr} = \text{Mod}_{1\text{Pr}}(1\text{Pr}), \quad X \longmapsto \mathbb{K}_{D,X},$$

valued one categorical degree above $D = D^{(1)}$.

Now the construction simply repeats, but now in Stefanich’s higher-categorical sense rather than by a literal replacement inside ordinary Pr^L . Once $D^{(2)}$ has been upgraded to a 2Pr -valued six-functor formalism, the same kernel construction can be performed one categorical level up: restricting $D^{(2)}$ to C/X and taking modules inside 2Pr produces the level-3 datum

$$\mathbb{K}_{D^{(2)},X} := \text{Mod}_{D^{(2)}}(\text{Fun}(\text{Span}(C/X), 2\text{Pr})) \in \text{CAlg}(3\text{Pr}),$$

and Proposition 3.1, applied to $D^{(2)}$, supplies its base-change-compatible adjoints, so $X \mapsto \mathbb{K}_{D^{(2)},X}$ upgrades in turn to a formalism $D^{(3)} : \text{Span}(C) \rightarrow 3\text{Pr}$. Inductively one obtains for every n a presentable six-functor formalism $D^{(n)} : \text{Span}(C) \rightarrow n\text{Pr}$, with $D^{(1)} = D$ and

$$D^{(n+1)}(X) = \mathbb{K}_{D^{(n)},X} := \text{Mod}_{D^{(n)}}(\text{Fun}(\text{Span}(C/X), n\text{Pr})) \in \text{CAlg}((n+1)\text{Pr}),$$

the same module construction one rung up — its coherence as an n -indexed tower being the content of Stefanich [Ste25]. Writing $A_n := D^{(n)}(X)$, the looping equivalences $A_n \simeq \text{End}_{A_{n+1}}(\mathbb{1})$ (Remark 3.2) assemble these levels into a *Stefanich ring*

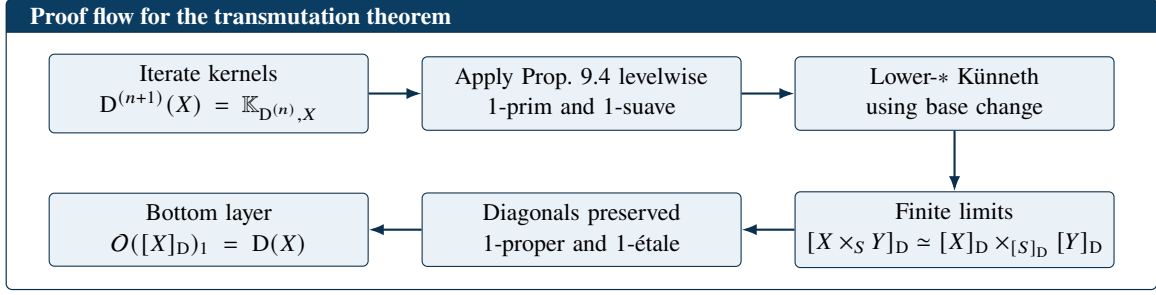
$$A_{D,X} = (D(X), \mathbb{K}_{D,X}, \mathbb{K}_{D^{(2)},X}, \dots) \in \text{StRing}, \quad \text{functorial in } X, \text{ i.e. a functor } C^{\text{op}} \rightarrow \text{StRing}.$$

This is the level- ∞ row of the ladder of categorical levels from the introduction: the single object that records every categorical degree of X at once.

Definition 3.3 (Transmutation). With $\text{Gest} := \text{StRing}^{\text{op}}$, the *transmutation* of D is

$$[-]_D : C \longrightarrow \text{Gest}_{/\text{Gest}(A_D)}, \quad X \longmapsto [X]_D = \text{Gest}(A_{D,X}).$$

The target is the slice over $\text{Gest}(A_D) = [*]_D$, the transmutation of the point; forgetting it reads $[-]_D$ simply as the functor $C \rightarrow \text{Gest}$ of the factorization diagram above.



Theorem 3.4 ([Sch26, Prop. 9.5]). *For every presentable six-functor formalism D , the transmutation $[-]_D$ satisfies:*

- (1) *it commutes with finite limits;*
- (2) *for each $f: Y \rightarrow X$ in C , the map $[f]_D: [Y]_D \rightarrow [X]_D$ is 1-étale and 1-proper ([Sch26, Def. 6.19 and 6.12]), with $f_{1\#} \simeq f_{1*}$ (the left and right adjoints to the kernel pullback $f_1^*: \mathbb{K}_{D,X} \rightarrow \mathbb{K}_{D,Y}$ agreeing);*
- (3) *the composite $O(-)_1 \circ [-]_D$ recovers the $(f^*, \otimes)$ -functoriality of D .*

Moreover the construction is, in Scholze's words, "completely canonical".

Proof. We separate the proof into four points.

First, we prove the adjointability statement. By construction, the Stefanich ring $A_{D,X}$ is obtained by iterating the kernel construction:

$$A_{D,X} = (D^{(1)}(X), D^{(2)}(X), D^{(3)}(X), \dots), \quad D^{(1)} = D, \quad D^{(n+1)}(X) = \mathbb{K}_{D^{(n)}, X}.$$

Thus the pullback on the m -th level of $A_{D,X}$ is exactly the kernel pullback for the $(m-1)$ -st sheaf theory $D^{(m-1)}$. Applying Proposition 3.1 to $D^{(m-1)}$, for every $m \geq 2$ we get left and right adjoints to

$$f_{m-1}^*: D^{(m)}(X) \longrightarrow D^{(m)}(Y),$$

and these adjoints agree:

$$f_{m-1,\#} \simeq f_{m-1,*}.$$

Moreover their formation is compatible with base change.

Unwinding the definitions of primness and suaveness for Stefanich rings, this is precisely the internal right-adjoint datum required for 1-primness and the internal left-adjoint datum required for 1-suaveness. The additional condition in the definition of n -suaveness is also satisfied. Indeed, the levelwise ambidexterity above lets us apply [Sch26, Lemma 6.22]: the adjoints preserve dualizable

objects. Since the unit is dualizable, the object $(B/A)_m^! = f_{m,\#}(\mathbb{1})$ is dualizable. For such a dualizable object, the extra right-adjoint condition in [Sch26, Def. 6.16] is equivalent to the already available left-adjoint condition. Hence every $[f]_{\mathcal{D}}$ is 1-prim and 1-suave.

Second, we prove that $[-]_{\mathcal{D}}$ commutes with finite limits. The terminal object is clear: the terminal object of $\mathcal{C}$ is sent to the base $[*]_{\mathcal{D}} = \text{Gest}(A_{\mathcal{D},*})$. It remains to prove preservation of pullbacks. After passing to a slice, it suffices to prove preservation of binary products. Fix maps

$$p_X: X \rightarrow S, \quad p_Y: Y \rightarrow S,$$

and put $P = X \times_S Y$, with projections $q_X: P \rightarrow X$ and $q_Y: P \rightarrow Y$. Let $p: P \rightarrow S$ be the structure map.

There is a canonical map of Stefanich rings over $A_{\mathcal{D},S}$:

$$\Phi: A_{\mathcal{D},X} \otimes_{A_{\mathcal{D},S}} A_{\mathcal{D},Y} \longrightarrow A_{\mathcal{D},P}.$$

It is induced by the two projection maps $P \rightarrow X$ and $P \rightarrow Y$. Thus multiplicativity and compatibility with the Stefanich tower are built into Φ ; we only have to check that its levelwise relative components are equivalences.

At categorical level n , the relative algebra of X over S is represented by

$$p_{X,n*} \mathbb{1}_X \in \mathcal{O}(A_{\mathcal{D},S})_n,$$

and similarly for Y and P . Therefore the level- n component of Φ is the canonical lower-star Künneth map

$$p_{X,n*} \mathbb{1}_X \otimes p_{Y,n*} \mathbb{1}_Y \longrightarrow p_{n*} \mathbb{1}_P.$$

This is a map of commutative algebra objects because it is the level- n component of the structured map Φ .

We now show that this map is an equivalence. The assignment

$$T \longmapsto \mathcal{O}(A_{\mathcal{D},T})_n$$

is a six-functor formalism with respect to the lower-star functors. This is exactly where the base-change part of Proposition 3.1 is used: without base change, the lower-star functors would not assemble functorially in cartesian squares. Using the projection formula and base change for the square defining $P = X \times_S Y$, we compute

$$p_{X,n*} \mathbb{1}_X \otimes p_{Y,n*} \mathbb{1}_Y \simeq p_{X,n*} (\mathbb{1}_X \otimes p_X^* p_{Y,n*} \mathbb{1}_Y)$$

and

$$p_X^* p_{Y,n*} \mathbb{1}_Y \simeq q_{X,n*} q_Y^* \mathbb{1}_Y = q_{X,n*} \mathbb{1}_P.$$

Hence

$$p_{X,n*} \mathbb{1}_X \otimes p_{Y,n*} \mathbb{1}_Y \simeq p_{X,n*} (\mathbb{1}_X \otimes q_{X,n*} \mathbb{1}_P) \simeq p_{X,n*} q_{X,n*} \mathbb{1}_P = p_{n*} \mathbb{1}_P.$$

Thus every level- n component of Φ is an equivalence. Since Φ is already a map of Stefanich rings, these equivalences are multiplicative and compatible with the tower. Therefore

$$A_{D,X} \otimes_{A_{D,S}} A_{D,Y} \simeq A_{D,P}.$$

Passing to opposites, this gives

$$[P]_D \simeq [X]_D \times_{[S]_D} [Y]_D.$$

Thus $[-]_D$ preserves pullbacks, hence finite limits.

Third, we upgrade 1-primness and 1-suaveness to 1-properness and 1-étaleness. In the Gestalt formalism, 1-properness means 1-primness for the map and for all its iterated diagonals, while 1-étaleness means 1-suaveness for the map and for all its iterated diagonals. Since we have just proved that $[-]_D$ preserves finite limits, it also preserves diagonals:

$$[\Delta_f]_D \simeq \Delta_{[f]_D}.$$

Applying the first part of the proof to f and to all iterated diagonals of f therefore gives that $[f]_D$ is 1-proper and 1-étale.

Fourth, the recovery statement is built into the definition of $A_{D,X}$. The first layer of the Stefanich ring is

$$O([X]_D)_1 = O(A_{D,X})_1 = D(X),$$

and the pullback and tensor functorialities on this first layer are exactly the original $(f^*, \otimes)$ -functoriality of D . Hence the composite

$$C^{\text{op}} \xrightarrow{[-]_D} \text{Gest}^{\text{op}} \xrightarrow{O(-)_1} \text{CAlg}(\mathbf{1Pr})$$

recovers the original functor D .

Finally, all constructions are canonical. The Stefanich ring $A_{D,X}$ is obtained functorially by iterating restriction to slices and the kernel category construction. The map Φ is induced by the universal property of the relative tensor product from the two projections $P \rightarrow X$ and $P \rightarrow Y$. The equivalences used above are the canonical projection-formula and Beck–Chevalley equivalences. Hence the resulting functor $X \mapsto [X]_D$ and its finite-limit comparison maps are canonical. $\square$

Warning 3.5. $O(-)_1 \circ [-]_D$ recovers D only at the level of $(f^*, \otimes)$; the shriek functors are not seen by the target $\text{CAlg}(\mathbf{1Pr})$. Whether the *full* formalism — the $!$ -functors included — can be reconstructed from $[-]_D$, by singling out a class of $!$ -able maps of Gestalten and a six-functor formalism on them, is the subject of the appendix to *Scholze’s Lecture IX* — not the present appendix — and is where Talk 11 takes over; the results there are, in Scholze’s words, “not definitive”.

4 Dictionary: six-functor notions vs. Gestalten

Having built $[-]_{\mathbf{D}}$, we can now read the geometric vocabulary of a map f directly off its image $[f]_{\mathbf{D}}$ in Gest: each six-functor property of f becomes a Gestalt-level property of $[f]_{\mathbf{D}}$.

By Theorem 3.4 every $[f]_{\mathbf{D}}$ is automatically 1-étale and 1-proper, with $f_{1\#} \simeq f_{1*}$; the categorical layers above the bottom are forced by ambidexterity and separate no maps. The whole dictionary therefore lives at layer 0, where the four Stefanich-ring conditions unwind directly to the six-functor notions of Section A. Concretely, the layer-0 datum of $[f]_{\mathbf{D}}$ is the unit map $\mathbb{1} \rightarrow (B/A)_1$ of the underlying map of Stefanich rings. This is a 1-morphism $[X] \rightarrow [Y]$ in $A_2 = \mathbb{K}_{\mathbf{D}, X}$, carrying the Fourier–Mukai kernel of f . Asking it to admit a right adjoint is 0-primness [Sch26, Def. 6.9]; asking a left adjoint is 0-suaveness [Sch26, Def. 6.16]. By Definition A.14 these say exactly that f is D-prim, resp. D-suave. Imposing the same on all iterated diagonals of f gives 0-properness and 0-étaleness [Sch26, Def. 6.12 and 6.19], which is the recursive Definition A.18; that recursion terminates only for truncated f , whence the extra truncatedness clause for the cohomological notions. All four equivalences then follow directly from the definitions.

Remark 4.1 ([Sch26, Rem. 9.6]). In tabular form:

f in C	$[f]_{\mathbf{D}}$ in Gest
suave	0-suave
prim	0-prim
cohomologically étale	truncated $\wedge$ 0-étale
cohomologically proper	truncated $\wedge$ 0-proper

Remark 4.2 (Stack-theoretic perspective). The suave and prim notions above have a clean stack-level form, due to Aoki [Aok26, Def. 4.5], stated in the absolute presentable kernel 2-category (with stacks the objects of its $\mathbf{CAlg}$): a morphism $[X] \rightarrow [Y]$ is *weakly suave* if it admits an $[X]$ -linear left adjoint, *prim* if it admits an $[X]$ -linear right adjoint, and *suave* if it is weakly suave and $[Y]$ is dualizable over $[X]$. For the kernel 2-category of a six-functor formalism, checking that $Y \rightarrow X$ in C determines such a morphism reduces to this $[X]$ -linear adjoint condition [Aok26, Example 4.8]. Aoki’s *weakly suave* matches the object-level Definition A.11 up to the standard reformulation “is a left adjoint” versus “admits an $[X]$ -linear left adjoint” (via the self-duality $\mathbb{K}_{\mathbf{D}, Y} \simeq \mathbb{K}_{\mathbf{D}, Y}^{\mathrm{op}}$), while his *suave* adds Aoki’s dualizability of $[Y]$ over $[X]$ — equivalently, by Observation A.16, of the suave dual — exactly the extra input promoting suaveness to cohomological smoothness. This is the stack-level reading behind the dictionary above: suaveness and primness of f become properties of $[f]_{\mathbf{D}}$ in Gest.

Outlook: towards the extension to stacks

One last question, and the most satisfying. The transmutation was built object by object — but does it see the *topology*? Tracking covers through $[-]_D$ reveals where the D-topology really comes from. Recall from [Sch25, Def. 5.13] the D-topology on C : a family $\{f_i: Y_i \rightarrow X\}$ (with each f_i in E and forming a cover for the canonical topology) is a D-cover iff it moreover satisfies *universal !-descent*, i.e. after any base change $X' \rightarrow X$ the upper-shriek functor

$$D(X') \xrightarrow{\sim} \lim \left(\prod_i D(Y_i \times_X X') \rightrightarrows \prod_{i,j} D(Y_i \times_X Y_j \times_X X') \cdots \right)$$

is an equivalence.

Proposition ([Sch26, Prop. 9.7]). If $\{f_i: Y_i \rightarrow X\}$ satisfies universal !-descent, then $\coprod_i [Y_i]_D \rightarrow [X]_D$ is a cover (effective epimorphism) in Gest.

Corollary ([Sch26, Cor. 9.8]). Hence $[-]_D$ extends to a functor

$$\mathrm{Shv}_D(C)^{\aleph_1} \longrightarrow \mathrm{Gest}/[_*]_D, \quad X \longmapsto [X]_D,$$

from D-sheaves on C (with values in 1-étale Gestalten), commuting with countable colimits and finite limits.

The upshot is that $[-]_D$ carries the D-topology on C into the canonical topology on Gest — every D-cover becomes a cover there — so in distilling a Grothendieck topology from a six-functor formalism one was already seeing, if only in secret, the canonical topology on Gestalten. Constructing the six-functor formalism *on* Gest itself — and pinning down the sense in which D can be recovered from it — is where Talk 11 takes over.

One-line summary

$D \rightsquigarrow \mathbb{K}_D \rightsquigarrow A_{D,X} \rightsquigarrow [X]_D = \mathrm{Gest}(A_{D,X})$: the $(f^*, \otimes)$ -part of the six-functor formalism is quasicohherent sheaves on its transmutation, canonically and for all D (the !-functors are another matter — see Warning 3.5) — so Gest is the universal category of geometric objects carrying a six-functor formalism.

Acknowledgements

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A Six-functor formalisms, smoothness, and suaveness

A modern viewpoint, originating in Grothendieck’s school, organizes “cohomology with coefficients” on a class of geometric objects C as a single coherent package: an assignment $X \mapsto D(X)$ of a category of “coefficient systems” to each object, together with six functors relating these categories under morphisms.

An informal definition. A *six-functor formalism* on C assigns to each $X \in C$ a presentable symmetric monoidal stable category $D(X)$, and to each morphism $f: Y \rightarrow X$ in C four functors

$$f^* \dashv f_*: D(X) \rightleftarrows D(Y), \quad f_! \dashv f^!: D(Y) \rightleftarrows D(X),$$

together with the symmetric monoidal product $\otimes$ and internal hom $\underline{\mathrm{Hom}}$ on each $D(X)$, such that:

- f^* is colimit-preserving and symmetric monoidal, and $f_!$ is colimit-preserving and satisfies the projection formula;
- base change: for every pullback square in C ,

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ f' \downarrow & \lrcorner & \downarrow f \\ X' & \xrightarrow{g} & X \end{array}$$

the canonical maps $g^* f_! \rightarrow f'_! g'^*$ and $f^! g_* \rightarrow g'_* f'^!$ are equivalences;

- projection formula: $f_!(A \otimes f^* B) \simeq f_!(A) \otimes B$ for $A \in D(Y)$, $B \in D(X)$.

These data, with all their higher coherences, are most cleanly encoded as a 2-functor out of a span 2-category — the precise formulation we take up in Section A.1. First we record the geometric content of each functor.

The six functors carry distinct geometric content:

- f^* is *pullback*, f_* the *direct image*, $\otimes$ the *symmetric monoidal product* encoding cup products, and $\underline{\mathrm{Hom}}$ the right adjoint to $\otimes$.
- $f_!$ is the *exceptional / proper-supported pushforward*: it agrees with f_* when f is proper and with $j_\#$ (extension by zero) when f is an open immersion.
- $f^!$ is the *exceptional pullback*, encoding orientation and duality data; in many examples, for f smooth of relative dimension d , $f^! \simeq f^* \otimes \omega_f$ with ω_f the relative dualizing complex.

The three operations f_* , $f^!$, $\underline{\text{Hom}}$ are *right adjoints* of f^* , $f_!$, $\otimes$ respectively, hence determined by the others. Modern formulations therefore typically specify only the three left functors $(f^*, f_!, \otimes)$ (a so-called *three-functor formalism*) and recover the rest by adjunction.

Three representative examples are:

- *Locally compact Hausdorff topological spaces* (Heyer–Mann [HM24]; see also [Sch25, Lecture VII]): here X ranges over (condensed) topological spaces and the coefficient category $\mathcal{D}(X)$ is the derived category of sheaves of Λ -modules on X , for a fixed coefficient ring Λ .
- *ℓ -adic étale cohomology of schemes* [Sch25, Appendix to Lecture VII].
- *The motivic formalism SH* [Sch25, Lecture XI] — the universal $\mathbb{A}^1$ -invariant one, by Drew–Gallauer.

For pedagogical introductions complementing the present appendix, we recommend Scholze’s lecture notes [Sch25], Gallauer’s mini-course [Gal21] (the source of much of this section’s exposition), and the textbook treatment in Cisinski–Déglise [CD19]. The encoding via span 2-categories adopted here is closest to Mann–Scholze [Man22; Sch25] and Cnossen–Lenz–Linskens [CLL25], to which we turn in the next subsection.

A.1 Span categories and suitable decompositions

The economical encoding of a six-functor formalism, initiated by Liu–Zheng [LZ24] and brought to its present form by Mann [Man22] and Scholze [Sch25], packages all the operations $(f^*, f_!, \otimes)$ and their coherences into a single lax symmetric monoidal functor on a span category. Given a wide subcategory $E \subset C$ closed under composition, base change, and diagonals (the last is needed for the kernel 2-category below — equivalently the inclusion $C_E \subseteq C$ preserves fibre products, [HM24, Def. 2.1.1 and Rem. 2.1.2]), the *span category* $\text{Span}(C, E)$ has the same objects as C and morphisms $X \rightarrow Y$ given by spans

$$X \leftarrow Z \xrightarrow{e} Y, \quad e \in E,$$

composed by pullback; when C has finite products, $\text{Span}(C, E)$ inherits a symmetric monoidal structure induced by the finite products of C , with unit the terminal object. (This monoidal structure is *not* Cartesian: the tensor $X \otimes Y \simeq X \times Y$ is induced by the *product* of C , and is in general neither the categorical product nor the coproduct of $\text{Span}(C, E)$. Independently, $\text{Span}(C, E)$ is semiadditive; but its biproduct $X \oplus Y \simeq X \sqcup Y$ comes from the *coproduct* of C , a different operation from the tensor.) A *three-functor formalism* on (C, E) is then a lax symmetric monoidal functor

$$\mathcal{D}: \text{Span}(C, E) \longrightarrow \text{Cat},$$

encoding f^* (backward legs), $f_!$ (forward legs), and $\otimes$ (monoidal structure) in a single coherent package; the right adjoints f_* , $f^!$, $\underline{\text{Hom}}$ are recovered automatically and are therefore *properties* of D rather than additional data.

The challenge is that constructing a lax monoidal functor out of $\text{Span}(C, E)$ directly is difficult. In practice, one starts from a much simpler piece of data, namely a functor

$$D_0: C^{\text{op}} \longrightarrow \text{CAlg}(\text{Cat})$$

encoding only f^* and $\otimes$, and asks when it extends to $\text{Span}(C, E)$. The key input is a *suitable decomposition* of E in the sense of Mann [Man22, Definition A.5.9]: a pair of wide subcategories $I, P \subset E$ (thought of as “étale-like” and “proper-like” morphisms respectively) such that every $e \in E$ factors as $e = p \circ i$ with $i \in I$ and $p \in P$, subject to mild cancellation and truncation axioms. The extension theorem then reads as follows. Suppose $i^* = D_0(i)$ admits a left adjoint $i_{\#}$ and $p^* = D_0(p)$ admits a right adjoint p_* , both satisfying the expected base change and projection formulae — in particular the mixed Beck–Chevalley equivalence $i_{\#}p'_* \simeq p_*i'_{\#}$. Then D_0 admits an essentially unique extension to a three-functor formalism D , in which

$$f_! = p_* i_{\#} \quad \text{for any factorization } f = p \circ i.$$

Recently, Cnossen–Lenz–Linskens [CLL25] gave a conceptually streamlined account of this extension theorem. They enhance $\text{Span}(C, E)$ to a 2-category whose 2-morphisms encode the adjunction data of $i_{\#}$ and p_* in a fully coherent way, and they prove that the inclusion $C^{\text{op}} \hookrightarrow \text{Span}_2(C, E)_{P, I}$ is a universal (I, P) -biadjointable functor. In this setup, Mann’s extension theorem becomes a formal corollary; as a byproduct, the unit and counit witnessing $i_{\#} \dashv i^*$ are part of the structure of D , rather than non-canonical choices recoverable only after the fact.

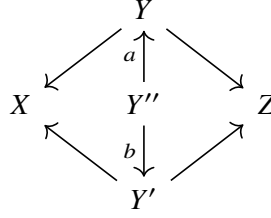
Construction A.1 (CLL 2-categorical span category). The 2-category $\text{Span}_2(C, E)_{P, I}$ has:

- *objects*: objects in C ;
- *1-morphisms* $X \rightarrow Z$: spans

$$\begin{array}{ccc} & Y & \\ f \swarrow & & \searrow g \\ X & & Z \end{array}$$

with $g \in E$;

- 2-morphisms between two such spans: diagrams



with $a \in P$ and $b \in I$.

Theorem A.2 ([CLL25, Theorem 5.17]). *Let C be a category with finite products and let $I, P \subseteq E \subseteq C$ be wide subcategories closed under base change such that*

- (1) *I and P are left cancellable;*
- (2) *every map in E factors as a map in I followed by a map in P ;*
- (3) *every map in $I \cap P$ is truncated.*

Let $D_0: C^{\text{op}} \rightarrow \text{CAlg}(\text{Cat})$ be a functor satisfying:

- (1) *for every $i: a \rightarrow b$ in I , the functor $i^*: D_0(b) \rightarrow D_0(a)$ admits a left adjoint $i_{\sharp}$ satisfying the Beck–Chevalley condition with respect to pullbacks in C and the projection formula $i_{\sharp}(- \otimes i^*(-)) \simeq i_{\sharp}(-) \otimes (-)$;*
- (2) *for every $p: a \rightarrow b$ in P , the functor $p^*: D_0(b) \rightarrow D_0(a)$ admits a right adjoint p_* satisfying the dual Beck–Chevalley condition and dual projection formula;*
- (3) *for every pullback square in C with vertical maps in P and horizontal maps in I , the mixed Beck–Chevalley map $j_{\sharp}p_* \rightarrow q_*i_{\sharp}$ is an equivalence.*

Then the lax symmetric monoidal functor $(C^{\times})^{\text{op}} \rightarrow \text{Cat}^{\times}$ corresponding to D_0 admits a unique extension to a lax symmetric monoidal 2-functor

$$D: \text{Span}_2(C, E)_{P, I}^{\otimes} \longrightarrow \text{Cat}^{\times}.$$

Moreover, for any $e \in E$ with factorization $e = p \circ i$ ($i \in I$, $p \in P$), the exceptional pushforward is $e_! \simeq p_ i_{\sharp}$.*

The above is the Mann-style extension formulation; equivalently, $C^{\text{op}} \hookrightarrow \text{Span}_2(C, E)_{P, I}$ is the universal (I, P) -biadjointable functor (left base change for I , right base change for P).

Thus, to construct a six-functor formalism it suffices to verify the three biadjointability conditions: left base change for I , right base change for P , and the mixed Beck–Chevalley equivalence. All coherences come for free from the universal property.

Throughout this subsection, (C, E) denotes a geometric setup and $D: \text{Span}(C, E) \rightarrow \text{Cat}$ a three-functor formalism (or a six-functor formalism, when stated). Following [HM24] we write Cat for the 2-category of categories and Cat_2 for the 2-category of 2-categories.

Cohomological smoothness

With the formalism in hand, we turn to the geometric properties of a morphism that D can detect, beginning with smoothness. The motivating example is Poincaré duality.

Theorem A.3 (Poincaré duality, classical). *Let $f: X \rightarrow Y$ be a submersion of topological manifolds of relative dimension d (i.e. locally of the form $Y \times \mathbb{R}^d \rightarrow Y$). Then $f_!$ admits a right adjoint $f^!$, and there is a sheaf $\omega_f := f^!(\mathbb{1}_Y) \in D(X, \mathbb{Z})$, locally isomorphic to $\mathbb{Z}[d]$, with*

$$f^! \simeq f^* \otimes \omega_f.$$

The sheaf ω_f is the dualizing complex of f . (For f proper one has instead $f_! \simeq f_$; the two phenomena are independent.)*

A morphism in E which satisfies the analogue of Poincaré duality in D will be called *cohomologically smooth* (or *D-smooth*). Two requirements are immediate:

- There is an isomorphism $f^! \simeq f^* \otimes \omega_f$ for some sheaf ω_f ;
- $\omega_f := f^!(\mathbb{1}_Y)$ is invertible.

But this is not yet enough. Consider a pullback square

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ g' \downarrow & \lrcorner & \downarrow g \\ X & \xrightarrow{f} & Y. \end{array}$$

We expect f' to be D-smooth as well, and the dualizing complexes to be compatible with restriction:

$$g'^* \omega_f \simeq \omega_{f'}.$$

This compatibility is not automatic from pure category theory; it must be imposed.

Definition A.4 (Cohomologically smooth morphism). A morphism $f: X \rightarrow Y$ in E is *cohomologically smooth* (or *D-smooth*) if it satisfies:

(1) The natural transformation

$$f^!(\mathbb{1}_Y) \otimes f^*(-) \longrightarrow f^*(-)$$

is an equivalence; we write $\omega_f := f^!(\mathbb{1}_Y)$ and call it the *dualizing complex* of f .

(2) The dualizing complex ω_f is $\otimes$ -invertible in $D(X)$.

(3) For every pullback square as above with $g \in E$, the morphism f' also satisfies (1)–(2), and the canonical map $g'^* f^!(\mathbb{1}_Y) \rightarrow f'^!(\mathbb{1}_{Y'})$ is an equivalence.

Remark A.5. Cohomological smoothness in the sense of Definition A.4 is weaker than the geometric notion of a smooth morphism. For instance, in the étale six-functor formalism $X \mapsto \mathrm{Shv}(X_{\text{ét}}, D_{\text{pf}}(\mathbb{Z}))$, every universal homeomorphism between schemes (a map of underlying topological spaces that remains a homeomorphism after pullback) is cohomologically smooth. The étale topology is insensitive to such maps, so they look like isomorphisms at the level of D — and isomorphisms are trivially D -smooth.

The kernel 2-category and Fourier–Mukai transforms

Verifying Definition A.4 directly is awkward because of the base-change clause (3). The next definitions package D -smoothness into a single 2-categorical statement, after which the base-change clause is automatic.

Fix $Y \in C$, and let $C_E \subseteq C$ be the wide subcategory whose morphisms are those in E . The slice $(C_E)_{/Y}$, paired with the class of *all* morphisms, is itself a geometric setup, and D restricts to a functor

$$D_Y: \mathrm{Span}((C_E)_{/Y}, \mathrm{all})^{\otimes} \rightarrow \mathrm{Span}(C, E)^{\otimes} \xrightarrow{D} \mathrm{Cat}^{\times}.$$

Definition A.6 (Fourier–Mukai transform). Given $X_1, X_2 \in (C_E)_{/Y}$ and $K \in D(X_1 \times_Y X_2)$, the *Fourier–Mukai transform with kernel K* is the functor

$$F_K: D(X_1) \rightarrow D(X_2), \quad A \longmapsto (\pi_2)_!(\pi_1^* A \otimes K).$$

A span $X_1 \xleftarrow{h} Z \xrightarrow{k} X_2$ over Y induces a Fourier–Mukai transform whose kernel is $K = (h, k)_! \mathbb{1}_Z$.

Remark A.7. The terminology is borrowed from the classical analytic transform $T: C^\infty(X) \rightarrow C^\infty(Y)$, $f \mapsto \int_X K(x, y) f(x) dx$, determined by its kernel K . Definition A.6 categorifies this construction: $\pi_1^* A$ pulls A back to a “function of two variables”, $\otimes K$ multiplies pointwise by the kernel, and $(\pi_2)_!$ is integration along the first variable.

Specializing to $f: X \rightarrow Y$ in E :

- Setting $X_1 = X$, $X_2 = Y$, and $K_1 = \mathbb{1}_X \in D(X \times_Y Y) = D(X)$, the Fourier–Mukai transform is $A \mapsto f_! A$.

- Setting $X_1 = Y$, $X_2 = X$, and $K_2 = \omega_f = f^!(\mathbb{1}_Y) \in D(Y \times_Y X) = D(X)$, the transform is $B \mapsto f^*B \otimes \omega_f$.

Idea A.8. The desired identity $f^! \simeq f^* \otimes \omega_f$ should arise by exhibiting the kernels K_1 and K_2 as an adjoint pair in a 2-category whose objects are schemes over Y and whose 1-morphisms are kernels. Since 2-functors preserve adjunctions, the resulting adjunction is automatically compatible with arbitrary base change $g: Y' \rightarrow Y$, taking care of clause (3) of Definition A.4 for free.

The invertibility of ω_f is a separate, relatively independent condition; this is what motivates the notion of suaveness below.

Definition A.9 (2-category of kernels). Let $Y \in C$. The 2-category of kernels $\mathbb{K}_{D,Y}$ is the 2-category whose:

- *Objects* are objects of $(C_E)_Y$, i.e., morphisms $X \rightarrow Y$ in E .
- *1-morphism category* from X_1 to X_2 is $\text{Fun}_Y(X_1, X_2) := D(X_1 \times_Y X_2)$.
- *Composition* of $M \in \text{Fun}_Y(X_1, X_2)$ and $N \in \text{Fun}_Y(X_2, X_3)$ is

$$N \circ M := (\pi_{13})_!(\pi_{12}^* M \otimes \pi_{23}^* N) \in D(X_1 \times_Y X_3),$$

where the π_{ij} are the projections from $X_1 \times_Y X_2 \times_Y X_3$.

The identity 1-morphism on X is $\Delta_f!(\mathbb{1}_X) \in D(X \times_Y X)$, where $\Delta_f: X \rightarrow X \times_Y X$ is the relative diagonal.

Conceptually, $\mathbb{K}_{D,Y}$ is obtained from $\text{Span}((C_E)_Y, \text{all})$, which is closed (being a span category over a category with finite limits) and hence self-enriched, by base-changing the self-enrichment along D_Y ; see [Aok26, Definition 2.4].

There is a canonical 2-functor

$$\Psi_{D,Y} = \text{Fun}_Y(Y, -) : \mathbb{K}_{D,Y} \longrightarrow \text{Cat},$$

sending an object X to $D(X)$ and a 1-morphism $M \in \text{Fun}_Y(X_1, X_2)$ to the Fourier–Mukai transform $(\pi_2)_!(\pi_1^*(-) \otimes M)$.

For the reader’s convenience, we record three facts about adjunctions in 2-categories that we use freely (proofs are standard, see [HM24, Appendix D]):

- Any 2-functor $F: \mathbb{C} \rightarrow \mathbb{D}$ preserves adjunctions; in fact, it preserves the entire mate correspondence [HM24, Lemma D.2.7].

- Adjunctions are unique up to a contractible space of choices: any two right adjoints to a given 1-morphism are canonically equivalent.
- A pair (f, g) with candidate unit η and counit ε such that the triangle composites are isomorphisms (not necessarily identities) can be corrected to a strict adjunction by modifying η .

Theorem A.10 (Smoothness via kernels, cf. [HM24, §4.5]). *A morphism $f: X \rightarrow Y$ in E is D-smooth if and only if*

- (1) *The 1-morphism $\mathbb{1}_X \in \mathrm{Fun}_Y(X, Y) = \mathrm{D}(X)$ is a left adjoint in $\mathbb{K}_{\mathrm{D}, Y}$. Denote its right adjoint by $\omega_f \in \mathrm{Fun}_Y(Y, X) = \mathrm{D}(X)$.*
- (2) *ω_f is $\otimes$ -invertible in $\mathrm{D}(X)$.*

When these hold, $\omega_f \simeq f^!(\mathbb{1}_Y)$.

Proof sketch. ($\Rightarrow$) Suppose f is D-smooth. By Definition A.4(1), $f^!(-) \simeq f^*(-) \otimes \omega_f$. Combined with the adjunction $f_! \dashv f^!$ this becomes $f_! \dashv f^*(-) \otimes \omega_f$ at the level of functors. Translating to $\mathbb{K}_{\mathrm{D}, Y}$, the kernel $\mathbb{1}_X$ corresponds to $f_!$ and ω_f to $f^*(-) \otimes \omega_f$, so $\mathbb{1}_X \dashv \omega_f$ in $\mathbb{K}_{\mathrm{D}, Y}$. (This lift of the functor-level adjunction to one in $\mathbb{K}_{\mathrm{D}, Y}$ rests on the explicit kernels above, not on any reflection property of $\Psi_{\mathrm{D}, Y}$, which is not conservative in general; cf. the Künneth obstruction of Proposition 2.3.) Invertibility is clause (2).

($\Leftarrow$) Apply $\Psi_{\mathrm{D}, Y}$ to the adjunction $\mathbb{1}_X \dashv \omega_f$. Since 2-functors preserve adjunctions, the resulting functors $\Psi_{\mathrm{D}, Y}(\mathbb{1}_X) \simeq f_!(-)$ and $\Psi_{\mathrm{D}, Y}(\omega_f) \simeq f^*(-) \otimes \omega_f$ form an adjunction in Cat_2 . By uniqueness of right adjoints, $f^*(-) \otimes \omega_f \simeq f^!(-)$, which is clause (1) of Definition A.4; setting $- = \mathbb{1}_Y$ gives $\omega_f \simeq f^!(\mathbb{1}_Y)$. The base-change clause (3) follows because $S \mapsto \mathbb{K}_{\mathrm{D}, S}$ assembles into a 2-functor $\mathrm{Span}(C, E)^\otimes \rightarrow \mathrm{Cat}_2^\times$ [HM24, Theorem 4.2.4], so the adjunction $\mathbb{1}_X \dashv \omega_f$ is preserved by base change. (The cited functor is lax symmetric monoidal; only its underlying 2-functoriality is used here.) $\square$

Suaveness and primness

Theorem A.10 decomposes cohomological smoothness as “left-adjointness in $\mathbb{K}_{\mathrm{D}, Y}$ ” plus “invertibility of the dualizing complex”. As foreshadowed, dropping the invertibility yields broader classes of objects.

Definition A.11 (Suave and prim objects). Let $f: X \rightarrow Y$ be in E and $A \in \mathrm{D}(X)$.

- (1) A is *f -suave* if, viewed as a 1-morphism $X \rightarrow Y$ in $\mathbb{K}_{\mathrm{D}, Y}$, it is a left adjoint. Its right adjoint is the *f -suave dual* $\mathrm{SD}_f(A) \in \mathrm{D}(X)$. Write $\mathrm{Suave}_f(X) \subseteq \mathrm{D}(X)$ for the full subcategory of f -suave objects.

- (2) A is f -*prim* if, viewed as a 1-morphism $Y \rightarrow X$ in $\mathbb{K}_{D,Y}$, it is a left adjoint. Its right adjoint is the f -*prim dual* $\mathrm{PD}_f(A) \in \mathrm{D}(X)$. Write $\mathrm{Prim}_f(X) \subseteq \mathrm{D}(X)$ for the full subcategory of f -prim objects.

(Equivalently, via the self-duality $\mathbb{K}_{D,Y} \simeq \mathbb{K}_{D,Y}^{\mathrm{op}}$, A is f -prim iff it is a *right* adjoint 1-morphism $X \rightarrow Y$; this is the form used by Heyer–Mann [HM24].)

Remark A.12 (Origin of the terminology). The terms are due to Scholze (suave) and Hansen (prim); applications typically require D to be a six-functor formalism, although the definition makes sense for three-functor formalisms.

- *Suave (relaxed smoothness)*: when $A = \mathbb{1}_X$ is f -suave, applying $\Psi_{D,Y}$ to the adjunction $\mathbb{1}_X \dashv \mathrm{SD}_f(\mathbb{1}_X)$ gives $f^*(-) \otimes \mathrm{SD}_f(\mathbb{1}_X) \simeq f^!(-)$. More generally, every f -suave A produces a relation between f^* and $f^!$.
- *Prim (primitive properness)*: when $A = \mathbb{1}_X$ is f -prim, the dual statement gives $f_!(\mathrm{PD}_f(\mathbb{1}_X) \otimes -) \simeq f_*(-)$. Primness is the phenomenon that f_* is recovered from $f_!$ by twisting; this is strictly weaker than the proper-base-change identity $f_! \simeq f_*$, so it is “primitive” in this sense.

In a precise sense, A is f -suave (or f -prim) iff it is “finite-dimensional relative to f ”: it admits a reflexive duality.

Proposition A.13 (Reflexive duality, [HM24, §4.4]). *Let D be a six-functor formalism on (C, E) and $f: X \rightarrow Y$ in E .*

- (1) *If $A \in \mathrm{D}(X)$ is f -suave, then $\mathrm{SD}_f(A)$ is also f -suave (with right adjoint A in $\mathbb{K}_{D,Y}$), and there is a natural isomorphism*

$$\mathrm{SD}_f(A) \otimes f^*(-) \simeq \underline{\mathrm{Hom}}(A, f^!(-)): \mathrm{D}(Y) \rightarrow \mathrm{D}(X).$$

Setting $- = \mathbb{1}_Y$ yields $\mathrm{SD}_f(A) \simeq \underline{\mathrm{Hom}}(A, f^!(\mathbb{1}_Y))$, exhibiting $\mathrm{SD}_f(A)$ as a relative Verdier dual of A . The bidual identity $\mathrm{SD}_f(\mathrm{SD}_f(A)) \simeq A$ then holds, and SD_f defines a contravariant equivalence

$$\mathrm{SD}_f: \mathrm{Suave}_f(X)^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Suave}_f(X).$$

- (2) *Dually, if $A \in \mathrm{D}(X)$ is f -prim, then $\mathrm{PD}_f(A)$ is also f -prim, with*

$$f_!(\mathrm{PD}_f(A) \otimes -) \simeq f_* \underline{\mathrm{Hom}}(A, -): \mathrm{D}(X) \rightarrow \mathrm{D}(Y),$$

and the bidual $\mathrm{PD}_f(\mathrm{PD}_f(A)) \simeq A$ holds. Consequently PD_f defines a contravariant equivalence

$$\mathrm{PD}_f: \mathrm{Prim}_f(X)^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Prim}_f(X).$$

Suave morphisms and prim morphisms

Specializing Definition A.11 to $A = \mathbb{1}_X$ yields classes of *morphisms*.

Definition A.14 (Suave and prim morphisms). Let $f: X \rightarrow Y$ be in E .

- (1) f is *D-suave* if $\mathbb{1}_X \in \text{Suave}_f(X)$. As before we write $\omega_f := \text{SD}_f(\mathbb{1}_X) \in \text{D}(X)$ for its *dualizing complex*.
- (2) f is *D-prim* if $\mathbb{1}_X \in \text{Prim}_f(X)$. We then write $\delta_f := \text{PD}_f(\mathbb{1}_X) \in \text{D}(X)$ for the *co-dualizing complex* of f .

Remark A.15. Combining Theorem A.10 with Definition A.14 gives

$$f \text{ is D-smooth} \iff f \text{ is D-suave and } \omega_f \text{ is invertible.}$$

By Proposition A.13, the dualizing complex is the relative Verdier dual $\omega_f \simeq f^!(\mathbb{1}_Y)$. The co-dualizing complex — the right adjoint in $\mathbb{K}_{\text{D},Y}$ of the unit kernel (Definitions A.9 and A.11) — is computed from the identity kernel by [HM24, §4.5]:

$$\delta_f \simeq \pi_{2*} \Delta_{f!} \mathbb{1}_X,$$

where $\pi_2: X \times_Y X \rightarrow X$ is the projection onto the second factor.

For D-suave f , the formula $\omega_f \otimes f^*(-) \xrightarrow{\sim} f^!(-)$ is equivalent (by adjunction) to $f^*(-) \xrightarrow{\sim} \underline{\text{Hom}}(\omega_f, f^!(-))$. Plugging in $\mathbb{1}_Y$ yields $\mathbb{1}_X \xrightarrow{\sim} \underline{\text{Hom}}(\omega_f, \omega_f)$, and for dualizable ω_f this gives $\mathbb{1}_X \simeq \omega_f^\vee \otimes \omega_f$. We record this as an observation.

Observation A.16. For a D-suave morphism f , the dualizing complex ω_f is $\otimes$ -invertible if and only if it is dualizable. Consequently the verification of D-smoothness reduces to checking suaveness and dualizability of ω_f .

We close with the base-change identities for suave and prim morphisms; they will guide the definitions of cohomologically étale and proper morphisms in the next subsection.

Lemma A.17 (Base change for suave and prim morphisms, [HM24, Lemma 4.5.13]). *Let D be a six-functor formalism on (C, E) and consider a pullback square*

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ g' \downarrow & \lrcorner & \downarrow g \\ X & \xrightarrow{f} & Y \end{array}$$

in C_E .

(1) If f is D-suave, the natural transformations

$$g'_* f'^* \simeq f^* g_*, \quad g'_! f'^! \simeq f^! g_!, \quad f'^* g'^! \simeq g'^! f^*, \quad g'^* f'^! \simeq f'^! g^*$$

are all isomorphisms.

(2) If f is D-prim, the natural transformations

$$g^* f_* \simeq f'_* g'^*, \quad g^! f_! \simeq f'_! g'^!, \quad f_! g'_* \simeq g_* f'_!, \quad f_* g'_! \simeq g'_! f_*$$

are all isomorphisms.

Cohomological étaleness and properness

We now ask: under what conditions are there genuine isomorphisms $f^! \simeq f^*$ and $f_! \simeq f_*$, rather than mere twists by ω_f or δ_f ?

A candidate map $f^! \rightarrow f^*$ is constructed using the diagonal. Consider

$$\begin{array}{ccccc} X & & \xrightarrow{\text{id}_X} & & X \\ & \searrow \Delta_f & & \searrow \downarrow & \\ & X \times_Y X & \xrightarrow{\pi_2} & X & \\ & \downarrow \pi_1 & \lrcorner & \downarrow f & \\ & X & \xrightarrow{f} & Y & \end{array}$$

id_X (curved arrow from X to X)

We have $f\pi_1 = f\pi_2$ and $\text{id}_X = \pi_2\Delta_f$. Suppose inductively that Δ_f satisfies $\Delta_f^! \simeq \Delta_f^*$. Then

$$\begin{aligned} f^! &\simeq \text{id}_X^* f^! \simeq \Delta_f^* \pi_2^* f^! \xrightarrow{\sim} \Delta_f^! \pi_2^* f^! \\ &\rightarrow \Delta_f^! \pi_1^! f^* \simeq (\pi_1 \Delta_f)^! f^* \simeq f^*, \end{aligned}$$

where $\pi_2^* f^! \rightarrow \pi_1^! f^*$ is induced (via right adjoints) from the base-change isomorphism $\pi_{1!} \pi_2^* \simeq f^* f_!$. By Lemma A.17, when f is D-suave the latter is an isomorphism, and so $f^! \simeq f^*$.

A dual construction gives $f_! \rightarrow f_*$, an isomorphism when Δ_f satisfies $\Delta_{f!} \simeq \Delta_{f*}$ and f is D-prim. To make the recursion well-founded we impose a truncatedness hypothesis: the recursion terminates at -2 -truncated morphisms (i.e., isomorphisms), where the conclusion is trivial.

Definition A.18 (Cohomologically étale and proper morphisms). Let D be a six-functor formalism on (C, E) , and let $f: X \rightarrow Y$ be a truncated morphism in E (i.e., n -truncated for some $n \in \mathbb{N}$).

- f is D-étale (or cohomologically étale) if f is D-suave and Δ_f is either D-étale or an isomorphism.

- f is *D-proper* (or *cohomologically proper*) if f is D-prim and Δ_f is either D-proper or an isomorphism.

These definitions are well-founded because n -truncated morphisms have $(n - 1)$ -truncated diagonals.

Proposition A.19 (Characterization of étaleness and properness, cf. [HM24, §4.6]). *Let D be a six-functor formalism on (C, E) and $f: X \rightarrow Y$ in E .*

(1) *Suppose Δ_f is D-étale. There is a natural map $f^! \rightarrow f^*$, and the following are equivalent:*

- (a) *f is D-étale;*
- (b) *$f^! \mathbb{1}_Y \simeq f^* \mathbb{1}_Y$ in $D(X)$;*
- (c) *$f^! \xrightarrow{\sim} f^*$ as functors.*

(2) *Suppose Δ_f is D-proper. There is a natural map $f_! \rightarrow f_*$, and the following are equivalent:*

- (a') *f is D-proper;*
- (b') *$f_! \mathbb{1}_X \simeq f_* \mathbb{1}_X$ in $D(Y)$;*
- (c') *$f_! \xrightarrow{\sim} f_*$ as functors.*

This completes the foundational vocabulary — cohomological smoothness, suaveness, primness, étaleness, properness — whose translation under the transmutation $[-]_D$ is the subject of the dictionary in the main text.

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